

Chapter 6

Wireless and

Mobile

~~Networks~~

A note on the

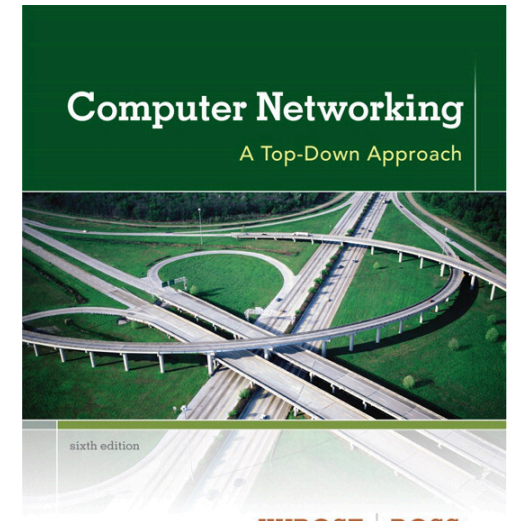
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orking: A
Top Down
Approach
6th edition
Jim Kurose, Keith
Ross
Addison-Wesley
March 2012

Ch. 6: Wireless and Mobile Networks

Background:

- ❖ # wireless (mobile) phone subscribers now exceeds # wired phone subscribers (5-to-

equal
devices

- laptops, Internet-enabled phones promise anytime untethered Internet access
- ❖ two important (but different) challenges
 - *wireless*: communication over wireless link
 - *mobility*: handling the mobile user who changes point of attachment to network

Chapter 6 outline

6.1 Introduction

Wireless

- CD

6.3 IEEE 802.11 wireless LANs (“Wi-Fi”)

6.4 Cellular Internet Access

- architecture
- standards (e.g., GSM)

Mobility

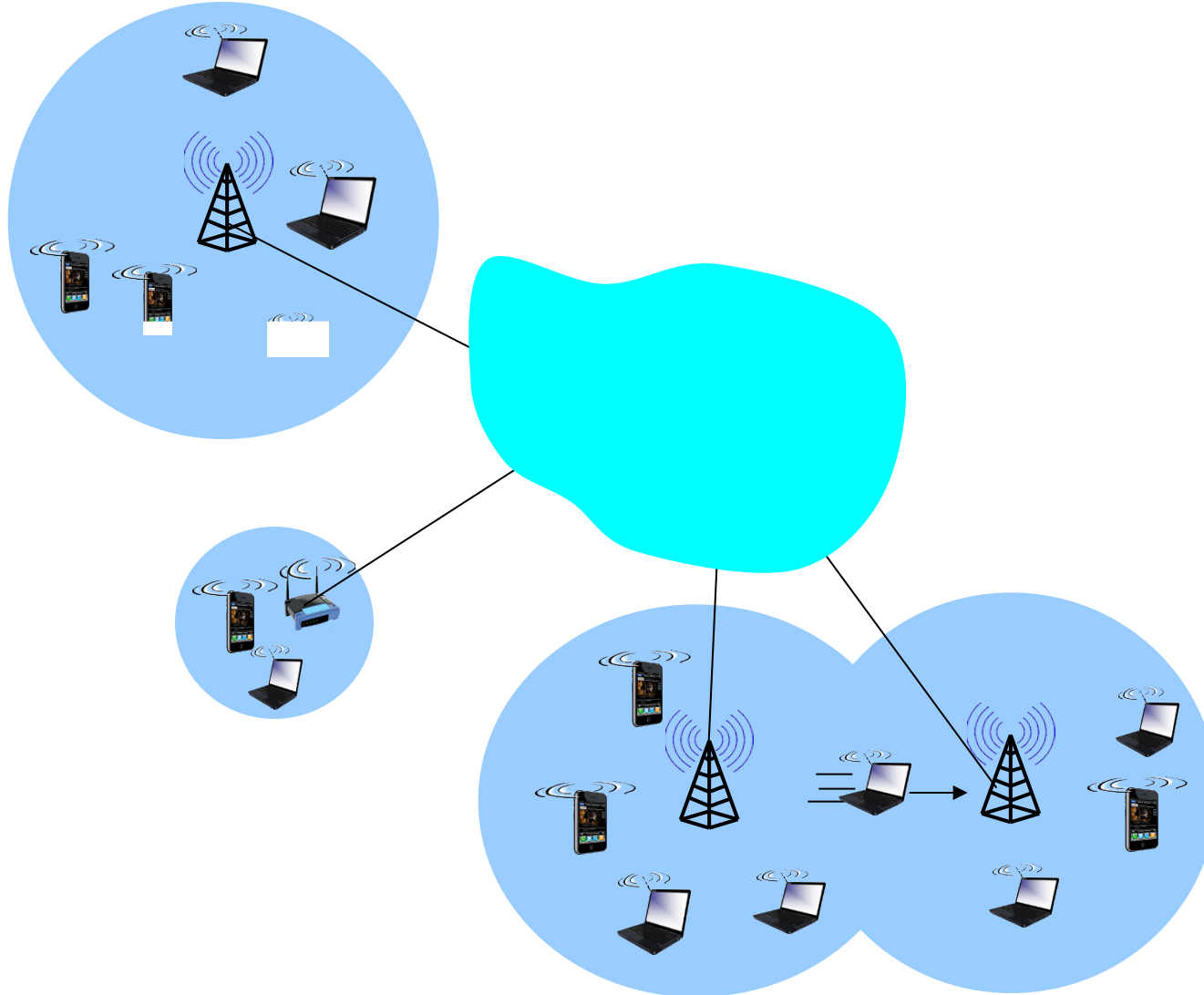
6.5 Principles: routing and

6.7 Handling mobility in cellular networks

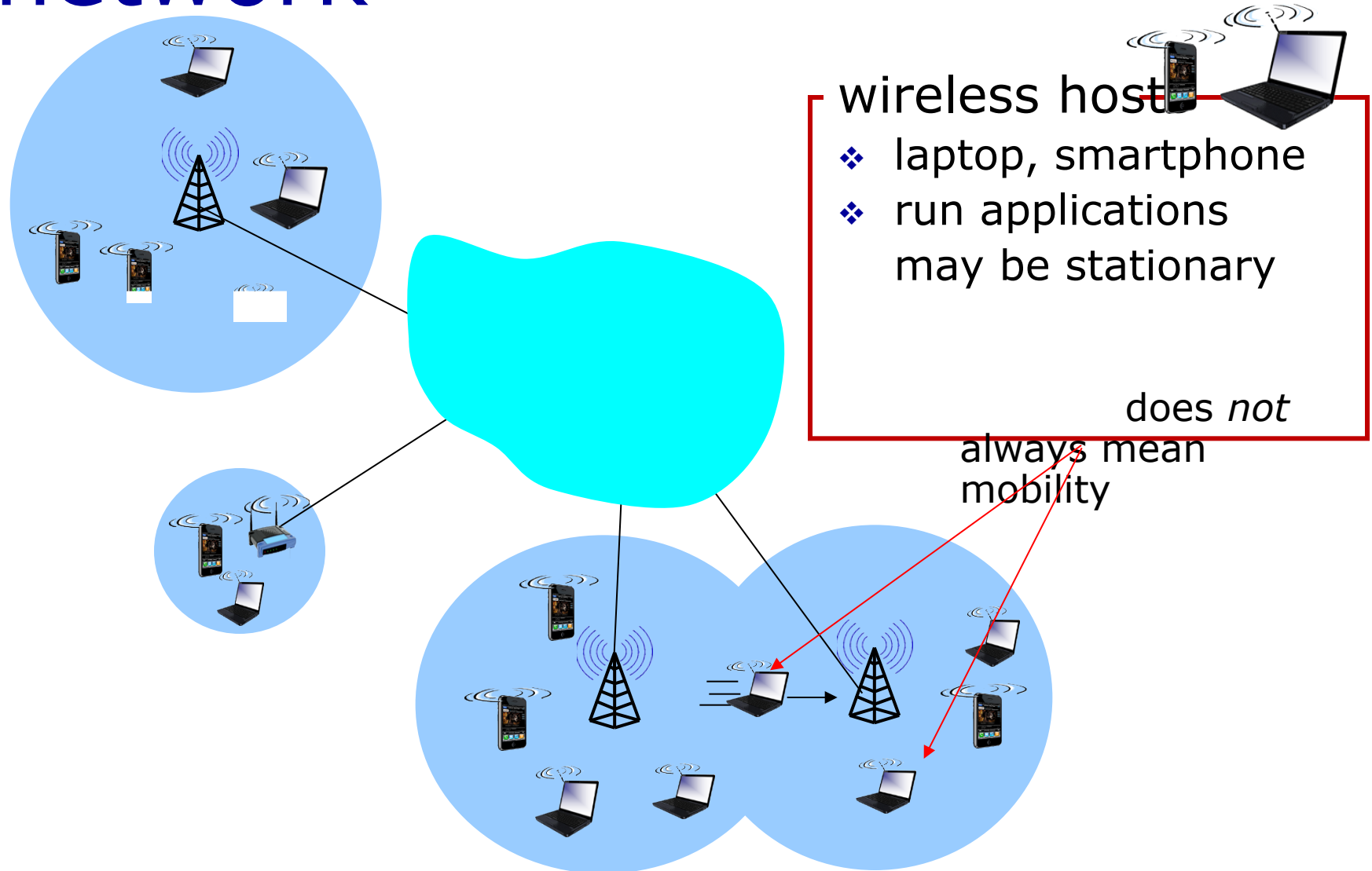
6.8 Mobility and higher- layer protocols

6.9 Summary

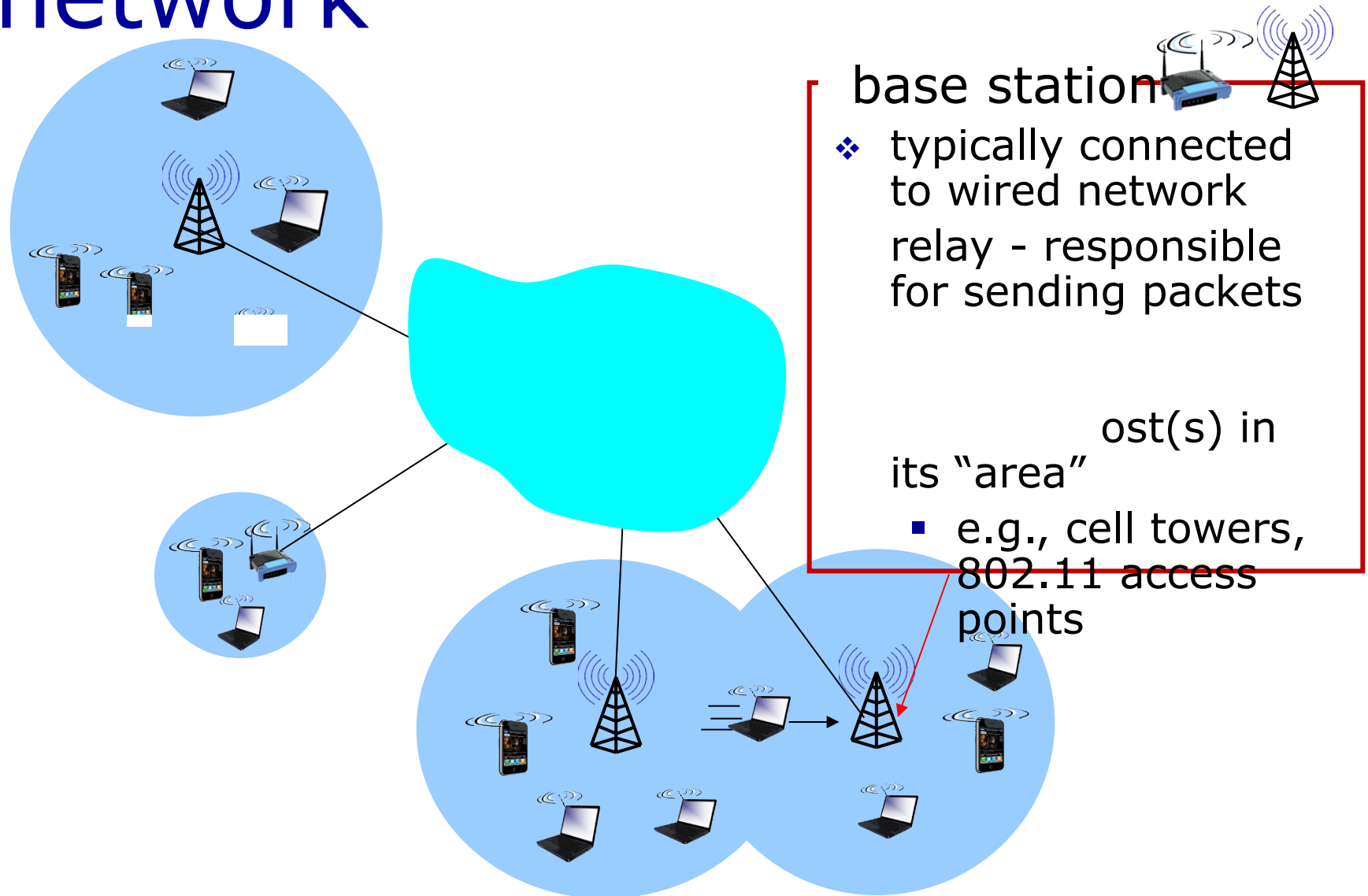
Elements of a wireless network



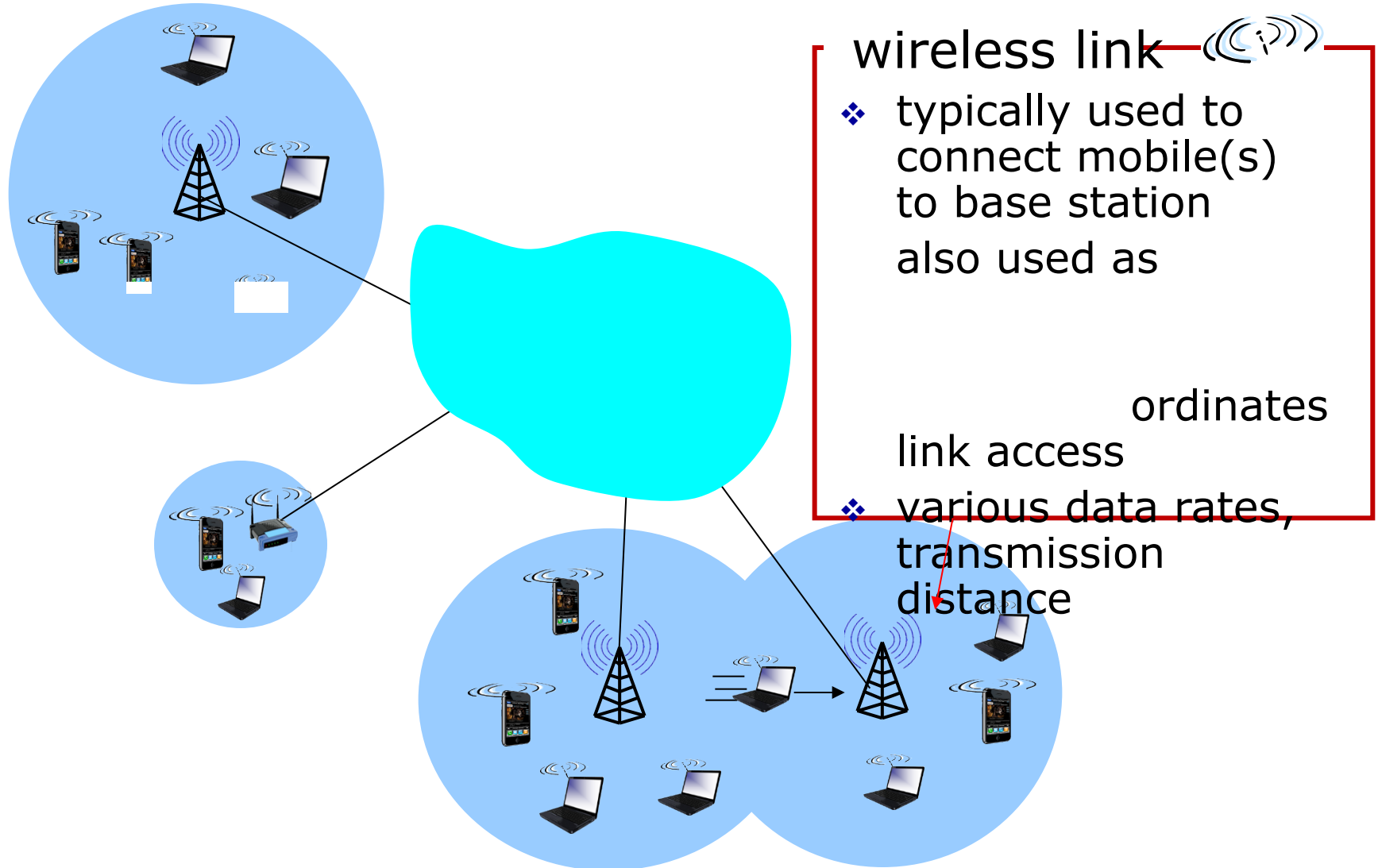
Elements of a wireless network



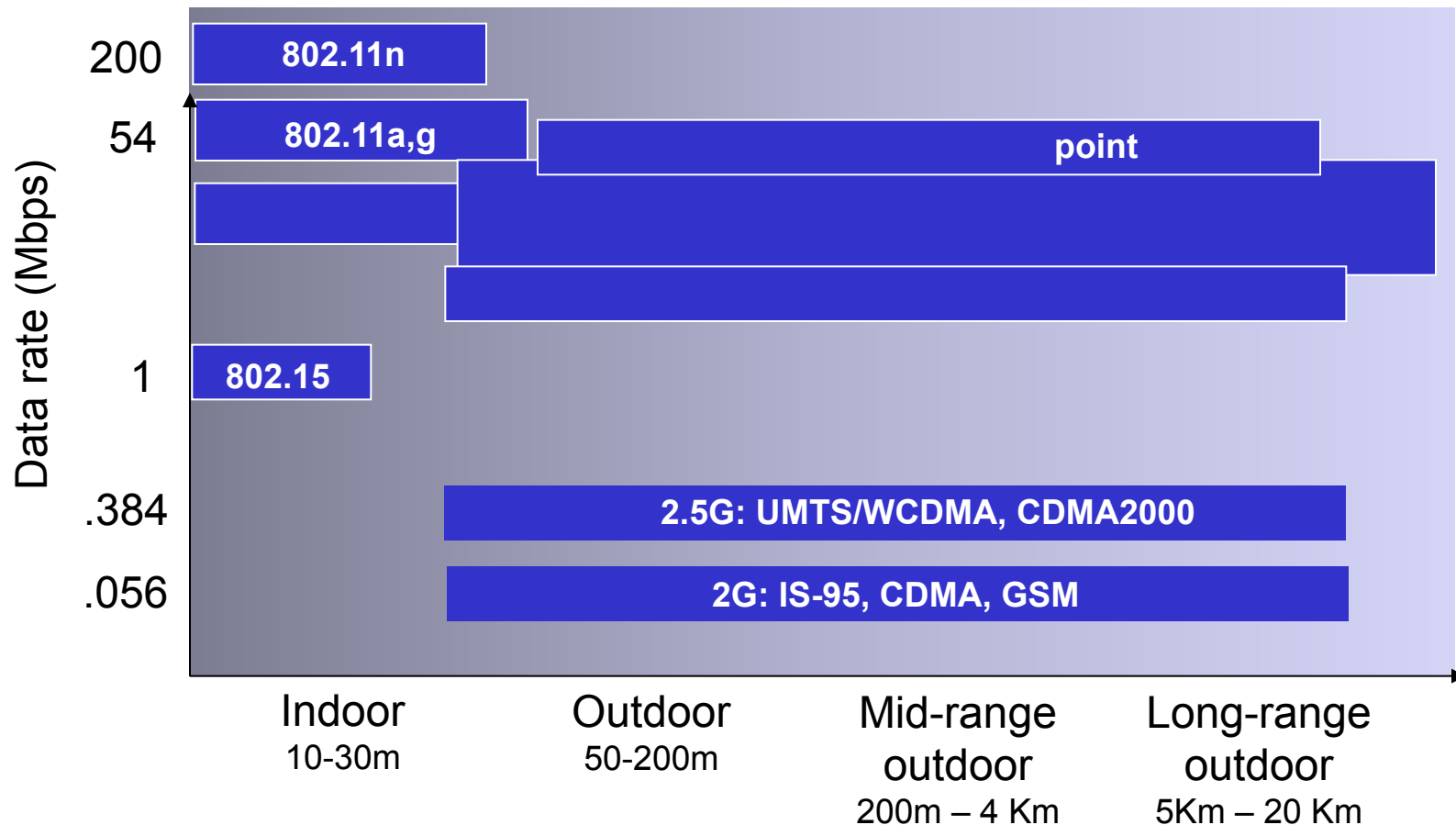
Elements of a wireless network



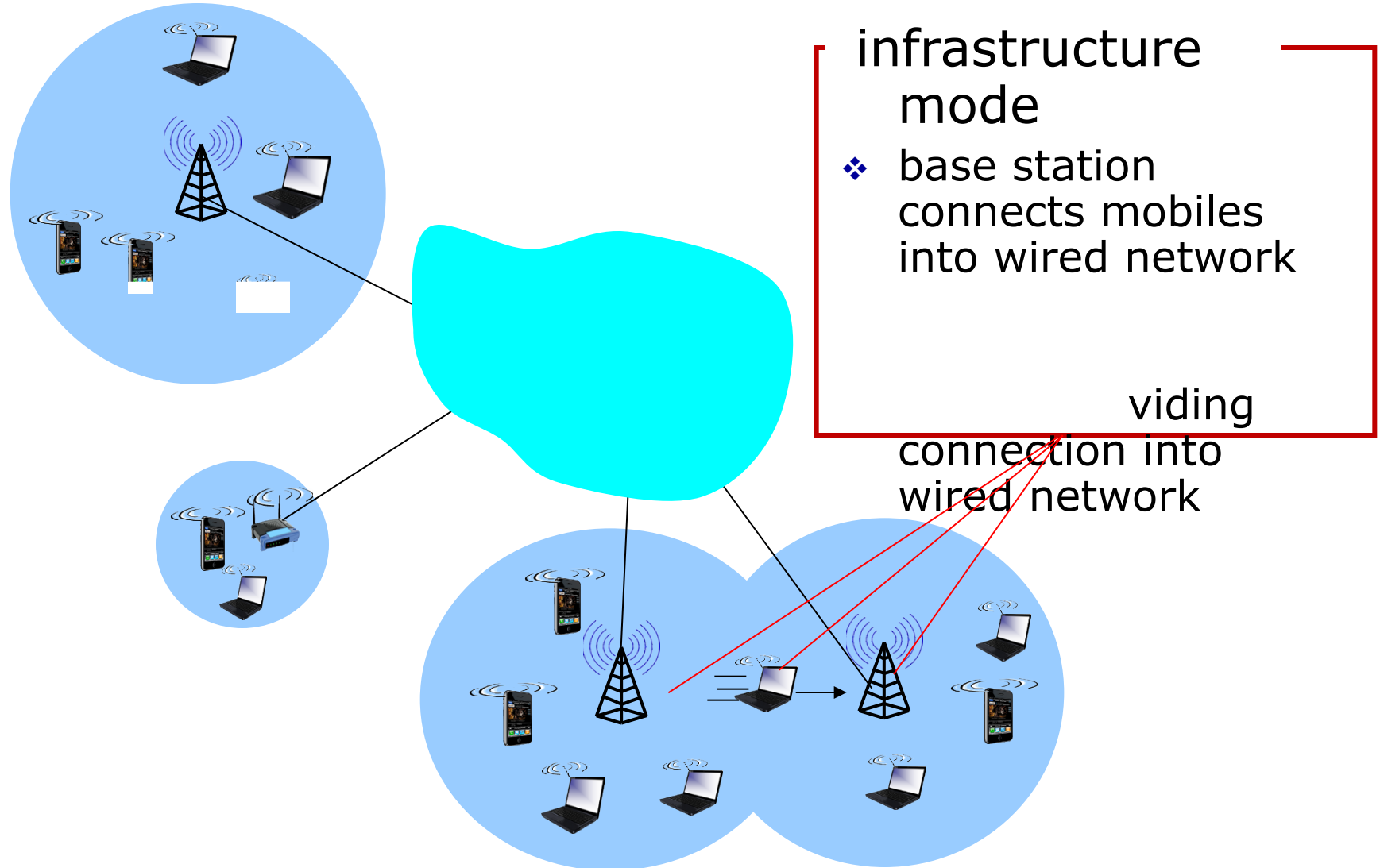
Elements of a wireless network



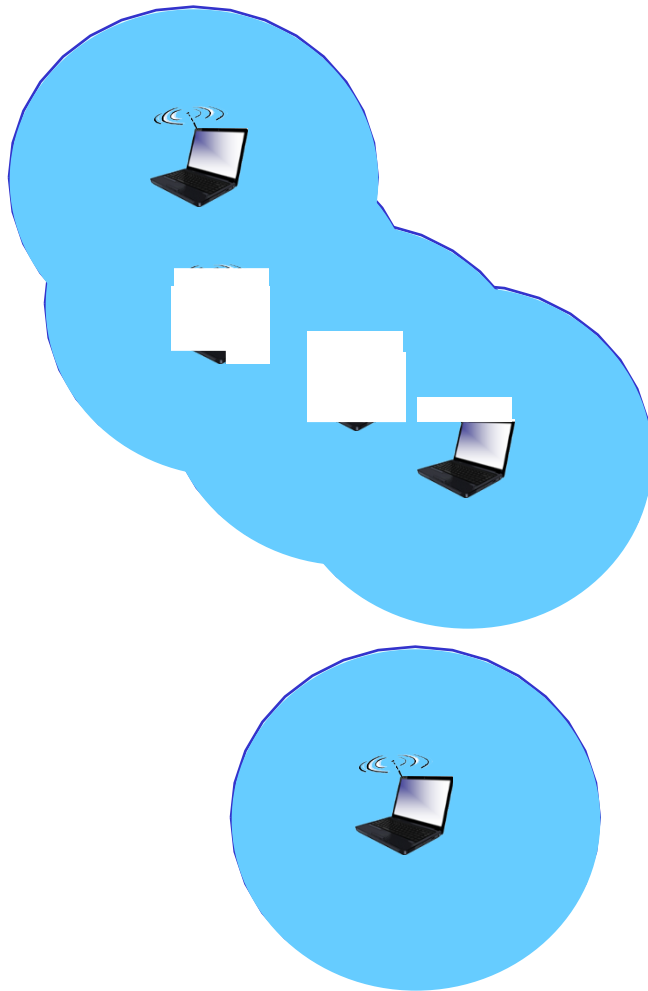
Characteristics of selected wireless links



Elements of a wireless network



Elements of a wireless network



ad hoc mode

- ❖ no base stations
- ❖ nodes can only transmit to other

organize
themselves into
a network: route
among
themselves

Wireless network ~~taxonomy~~

	single hop	multiple hops
infrastructure	host connects to base	host may have to through several
no infrastructure	no base station, no connection to larger Internet (Bluetooth, ad hoc nets)	no base station, no connection to larger Internet. May have to relay to reach other a given wireless node MANET, VANET

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Wireless Link Characteristics (1)

important differences from wired link

- *decreased signal strength*: radio signals attenuate through matter

interferences
(e.g., 2.4 GHz) shared by other devices
(e.g., phone); devices (motors) interfere as well

- *multipath propagation*: radio signal reflects off objects ground, arriving at destination at slightly different times

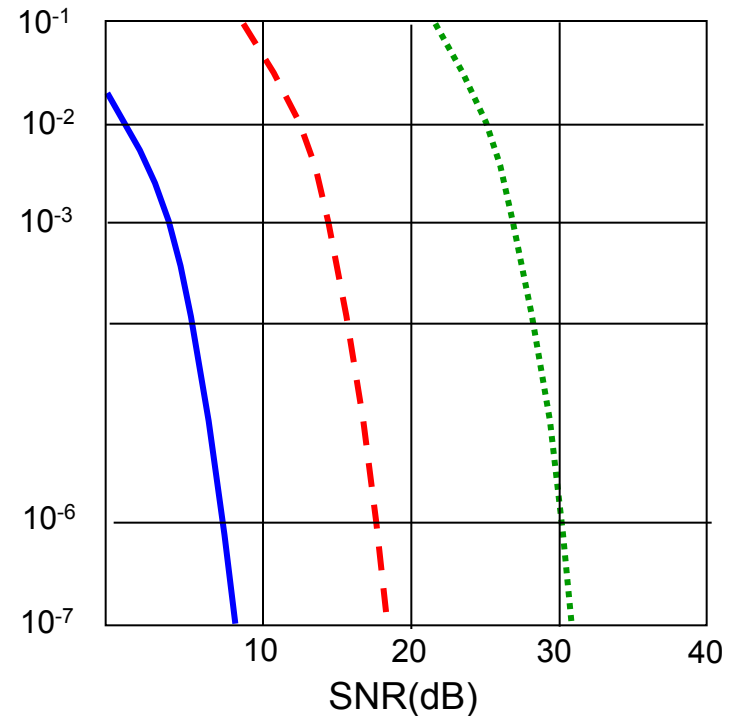
.... make communication across (even a point to point) wireless link much more “difficult”

Wireless Link Characteristics (2)

- ❖ SNR: signal-to-noise ratio
 - larger SNR – easier to extract signal

trade

- *given physical layer:* increase power -> increase SNR->decrease BER
- *given SNR:* choose physical layer that meets BER requirement, giving highest throughput
 - SNR may change with



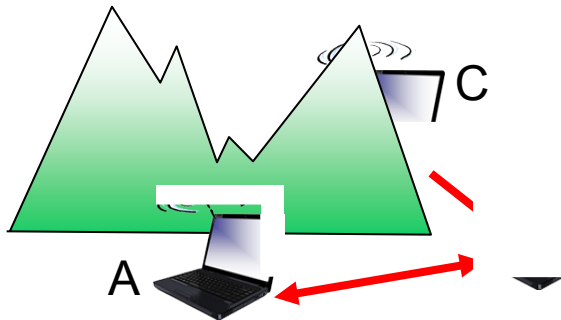
..... QAM256 (8 Mbps)

- - - QAM16 (4 Mbps)

— BPSK (1 Mbps)

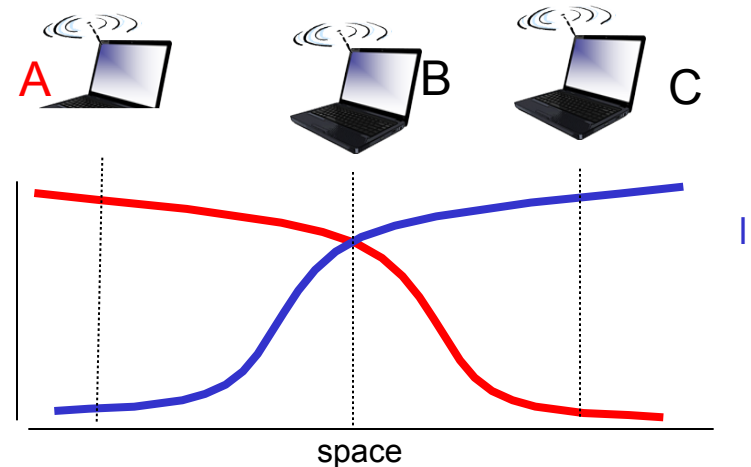
Wireless network characteristics

Multiple wireless senders and receivers create additional problems (beyond multiple access):



Hidden terminal problem

- ❖ B, A hear each other
- ❖ B, C hear each other
- ❖ A, C can not hear each other means A, C unaware of their interference at B



Signal attenuation:

- ❖ B, A hear each other
- ❖ B, C hear each other
- ❖ A, C can not hear each other other interfering at B

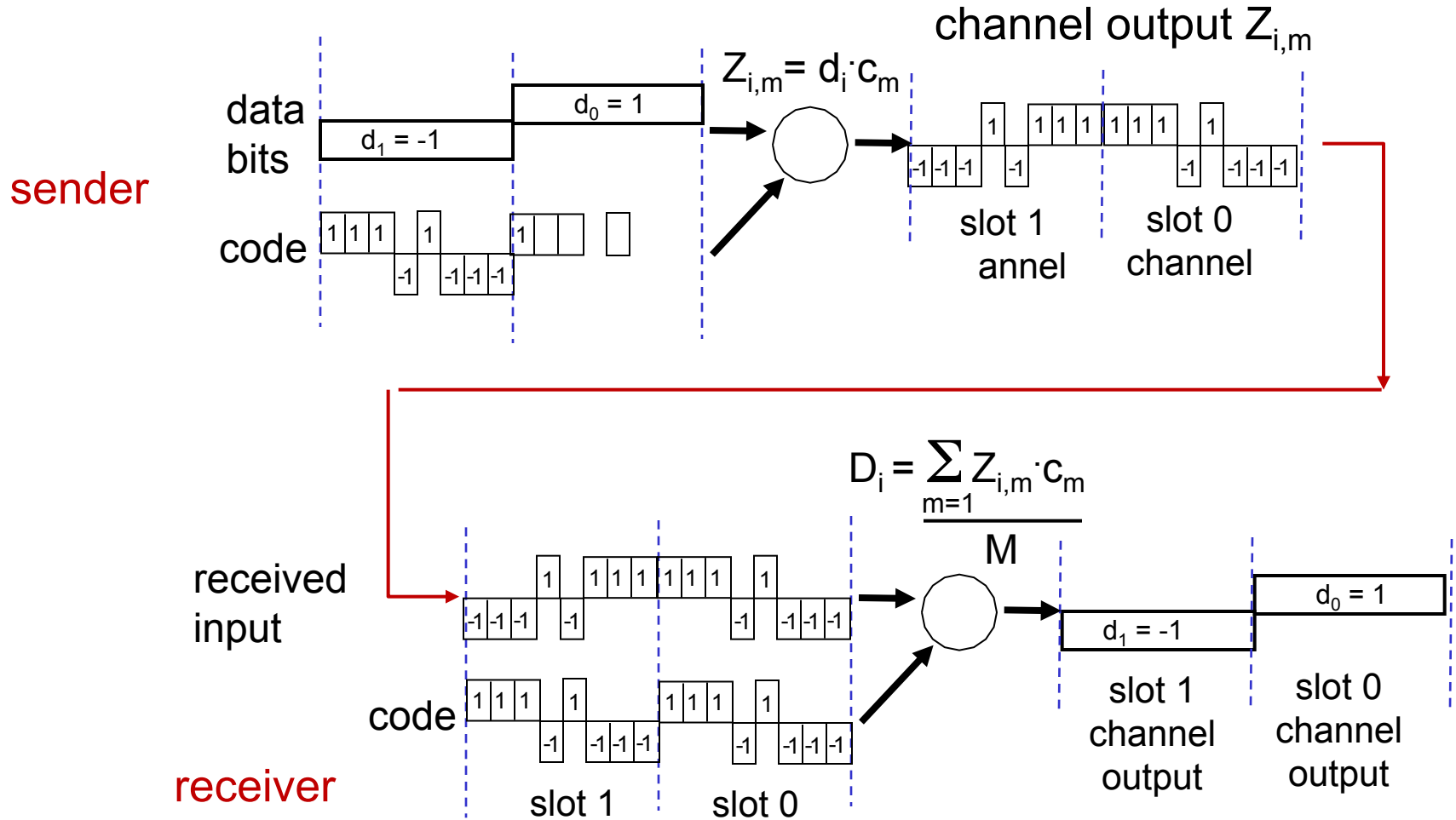
Code Division Multiple Access (CDMA)

- ❖ unique “code” assigned to each user;
i.e., code set partitioning
 - all users share same frequency, but each user has own code to encode (i.e.,

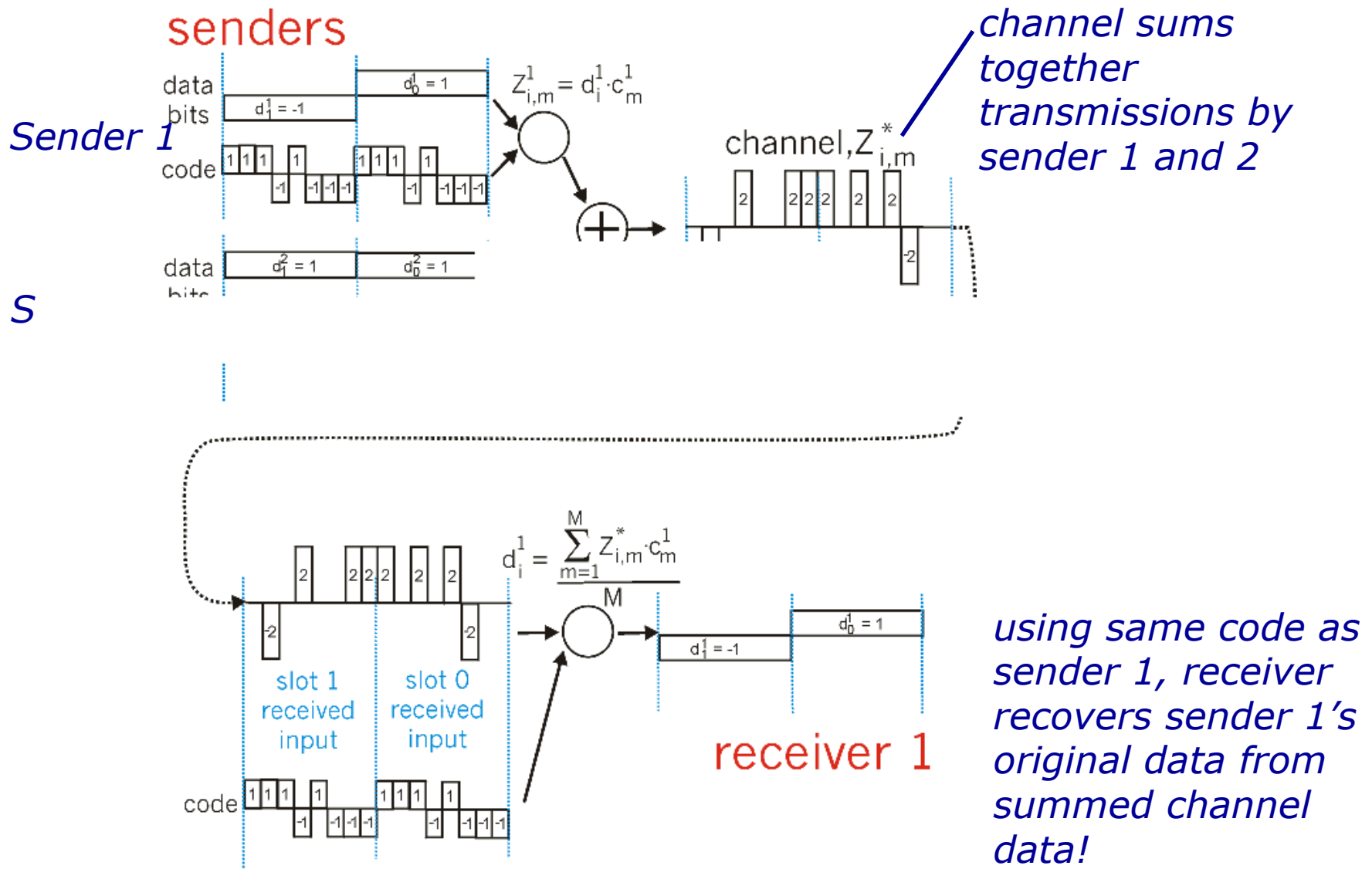
integrate

- ❖ *encoded signal* = (original data) X (chipping sequence)
- ❖ *decoding*: inner-product of encoded signal and chipping sequence

CDMA encode/decode



CDMA: two-sender interference



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IEEE 802.11 Wireless LAN

802.11b

- ❖ 2.4-5 GHz unlicensed spectrum



spectrum
physical layer

- all hosts use same chipping code

- ❖ all use CSMA/CA for multiple access
- ❖ all have base-station and ad-hoc network versions

802.11a

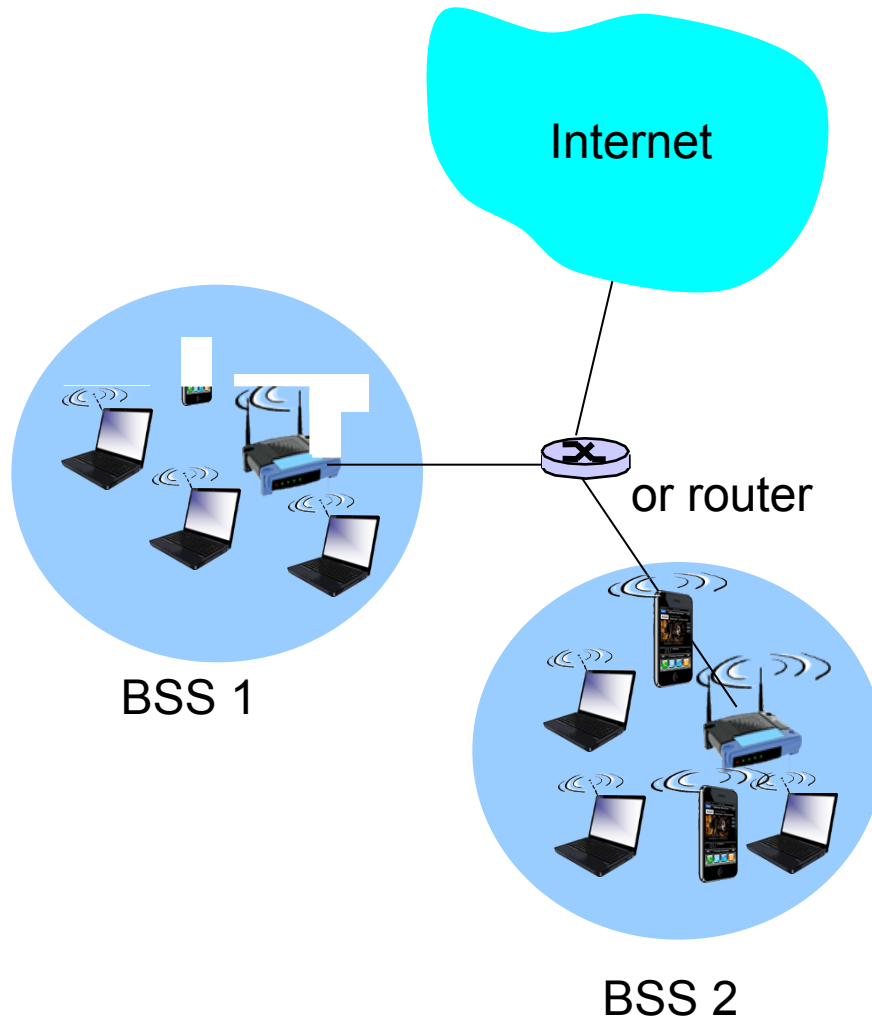
- 5-6 GHz range
- up to 54 Mbps

1g

802.11n: multiple antennae

- 2.4-5 GHz range
- up to 200 Mbps

802.11 LAN architecture



- ❖ wireless host communicates with base station
base station = access

cell") in infrastructure mode contains:

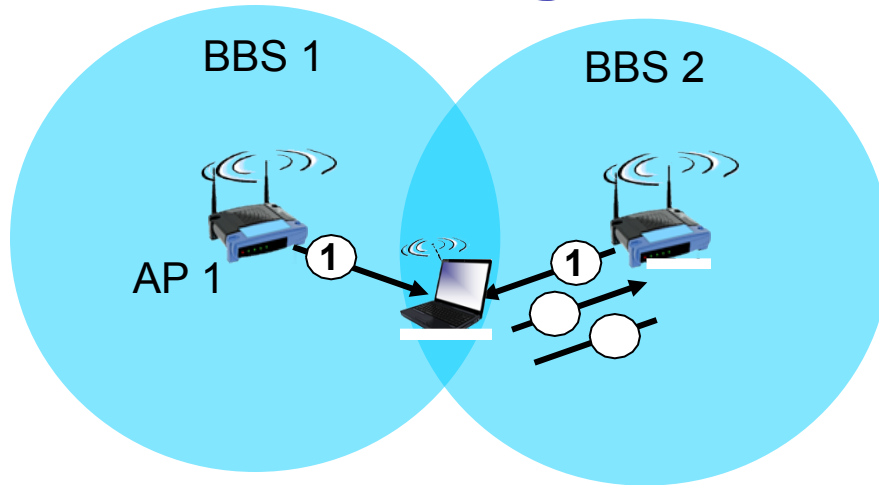
- wireless hosts
- access point (AP): base station
- ad hoc mode: hosts only

802.11: Channels, ~~association~~

- ❖ 802.11b: 2.4GHz-2.485GHz spectrum divided into 11 channels at different frequencies
 - AP admin ch for AP

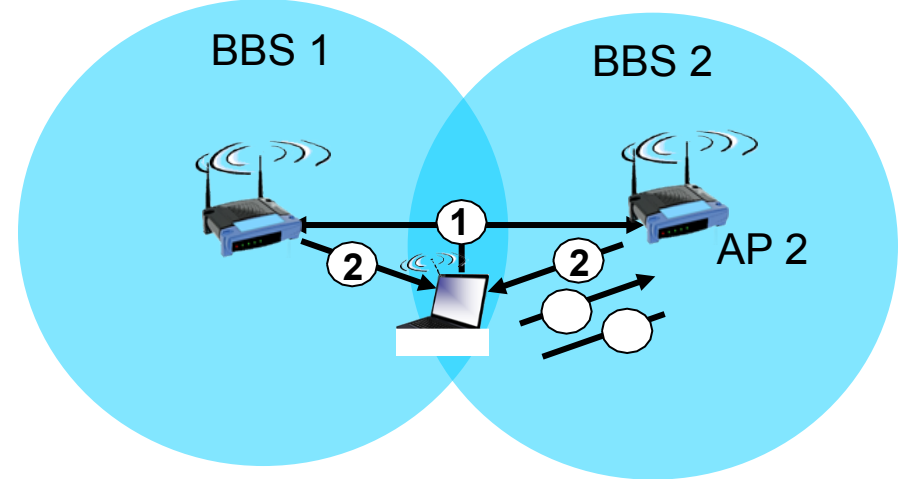
- ❖ host:
 - scans channels, listening for *beacon frames* containing AP's name (SSID) and MAC address
 - selects AP to associate with
 - may perform authentication [Chapter 8]
 - will typically run DHCP to get IP address in AP's subnet

802.11: ~~passive/active~~ scanning



passive scanning:

- (1) beacon frames sent from APs
- (2) association Request frame sent: H1 to selected AP
- (3) association Response frame sent from selected AP to H1



active scanning:

- (1) Probe Request frame broadcast from H1
- (2) Probe Response frames sent from APs
- (3) Association Request frame sent: H1 to selected AP
- (4) Association Response frame sent from selected AP to H1

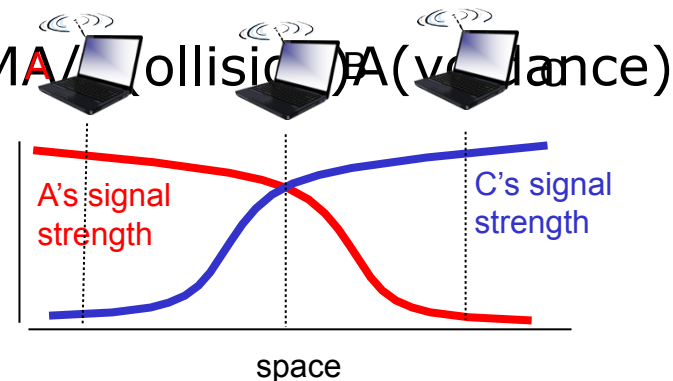
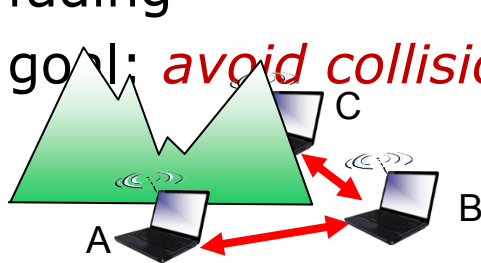
IEEE 802.11: multiple access

- ❖ avoid collisions: 2+ nodes transmitting at same time
- ❖ 802.11: CSMA - sense before transmitting
 - don't collide with transmission by other node

due

- can't sense all collisions in any case: hidden terminal, fading

- goal: *avoid collisions*: CSMA/CA (avoid collision)



IEEE 802.11 MAC Protocol: CSMA/CA

802.11 sender

1 if sense channel idle for **DIFS** then
transmit entire frame (no CD)

2 if sense channel busy t

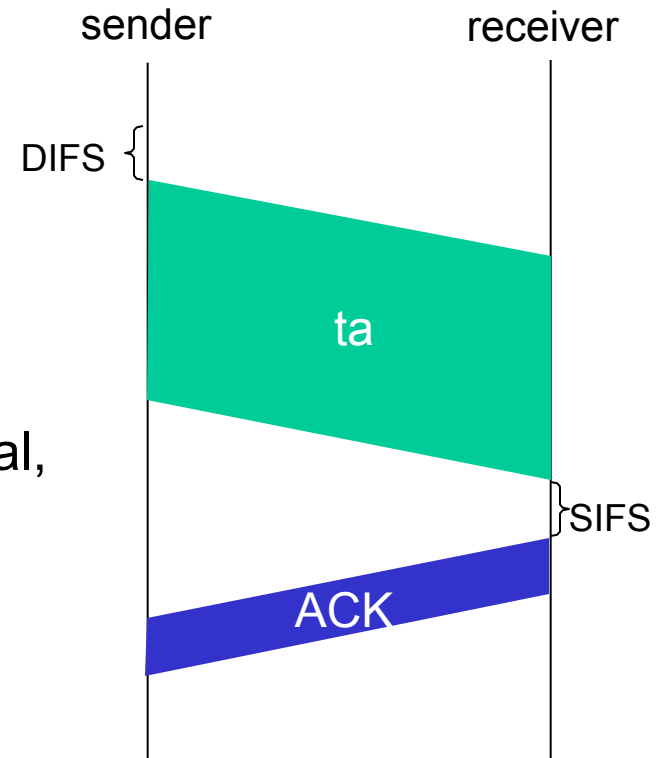
transmit

if no ACK, increase random backoff interval,
repeat 2

802.11 receiver

- if frame received OK

return ACK after **SIFS** (ACK needed due to
hidden terminal problem)



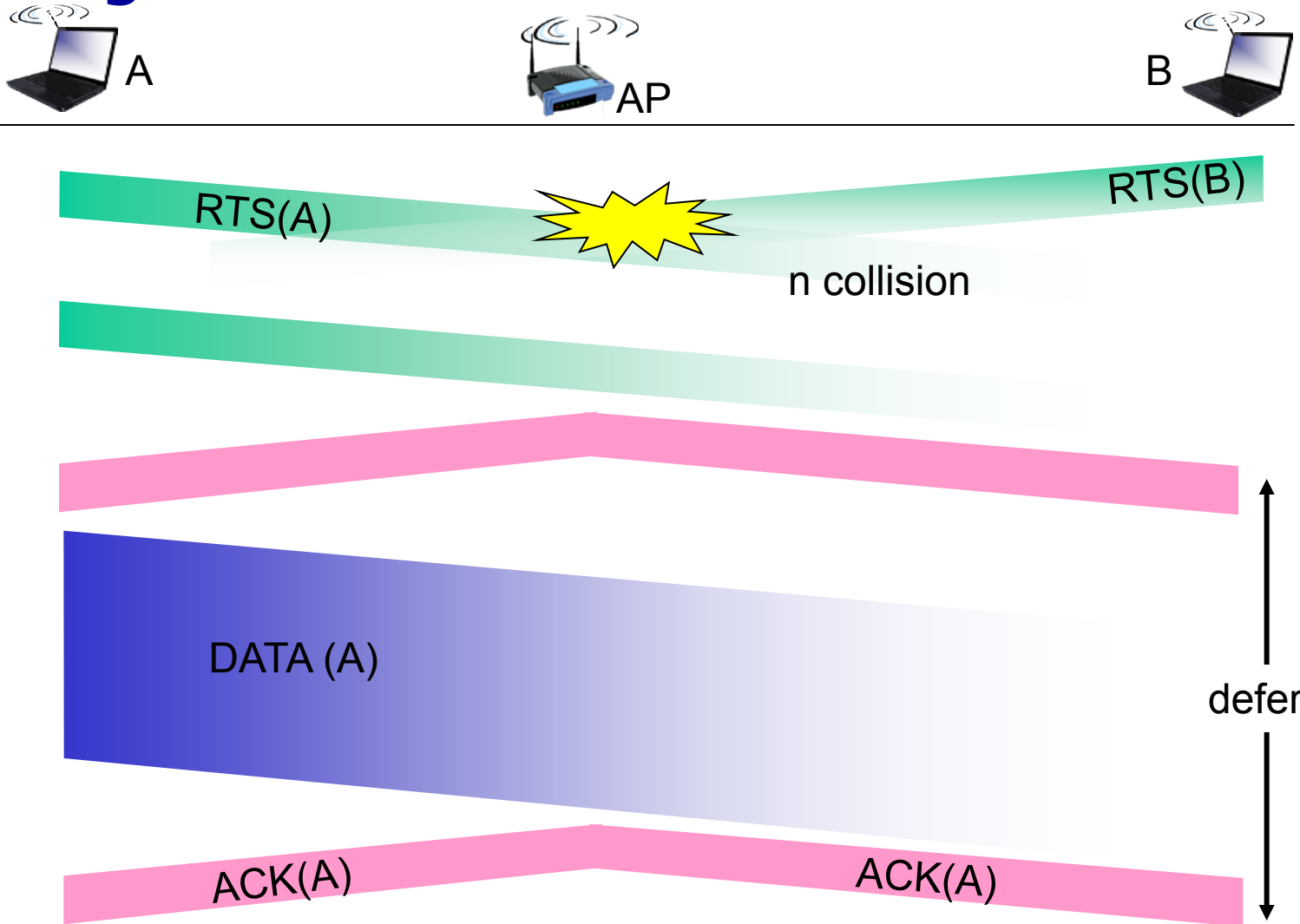
Avoiding collisions (more)

idea: allow sender to “reserve” channel rather than random access of data frames: avoid collisions of long data frames

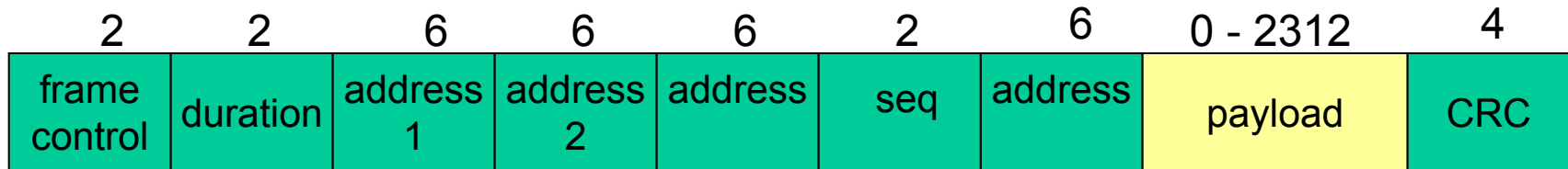
- RTS y’re
short)
- ❖ BS broadcasts clear-to-send CTS in response to RTS
- ❖ CTS heard by all nodes

avoid data frame collisions completely using small reservation packets!

Collision Avoidance: RTS-CTS exchange



802.11 frame: ~~addressing~~



Address 1:
of wireless host or AP
to receive this frame

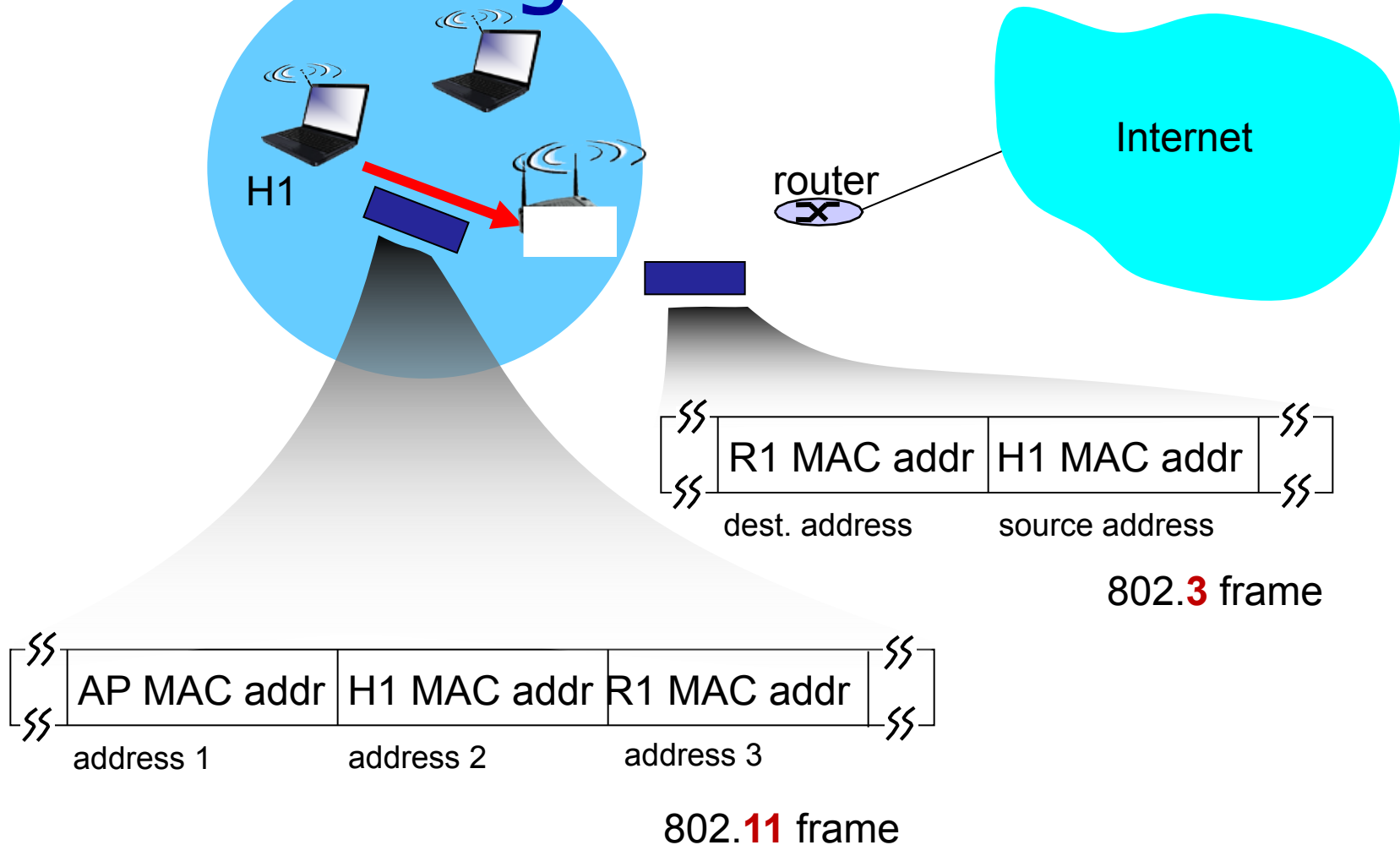
Address 2: MAC address
of wireless host or AP
transmitting this frame

Address 3: MAC
address
of router interface to
which AP is attached

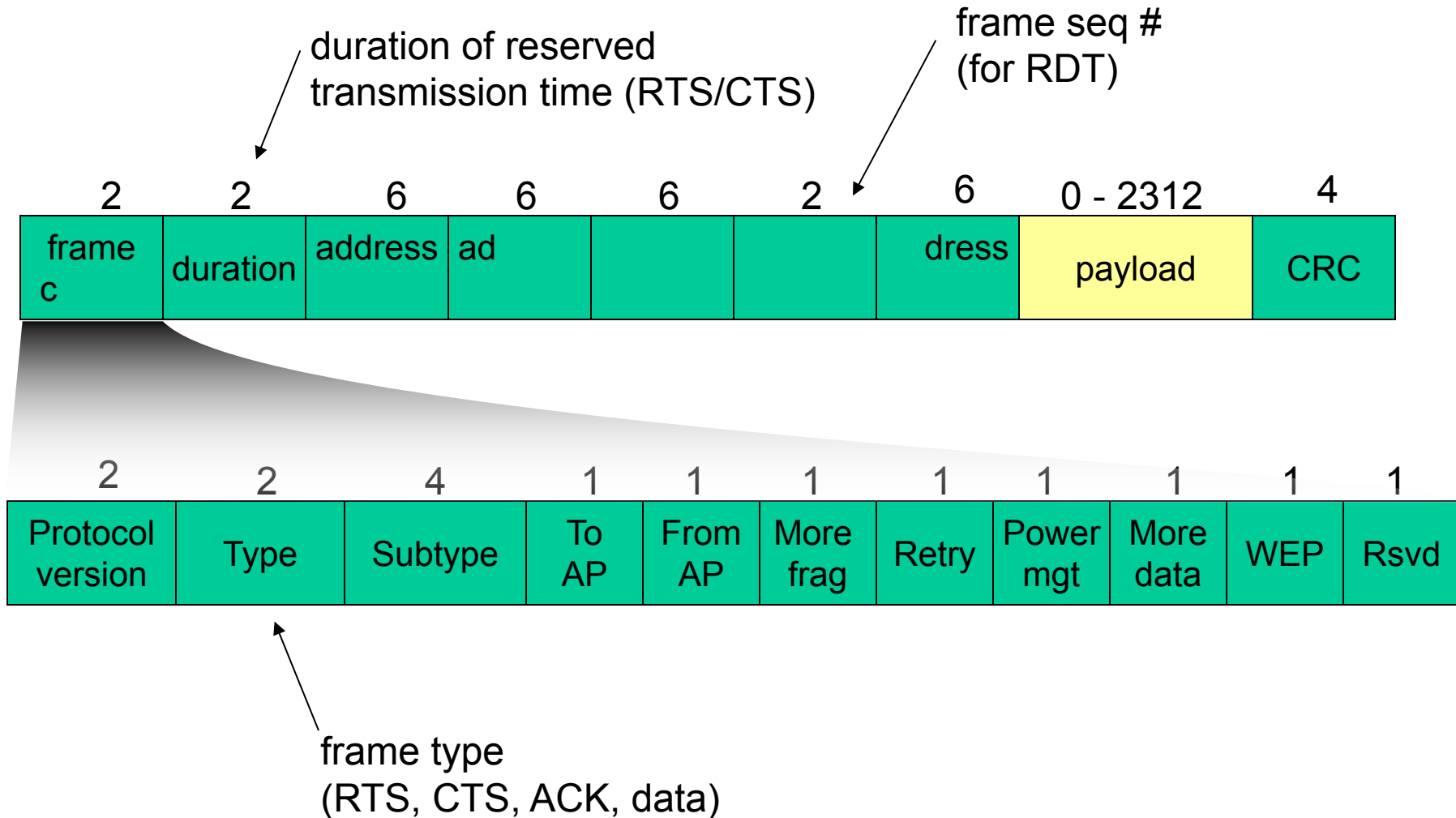
hoc

mode

802.11 frame: addressing



802.11 frame: more

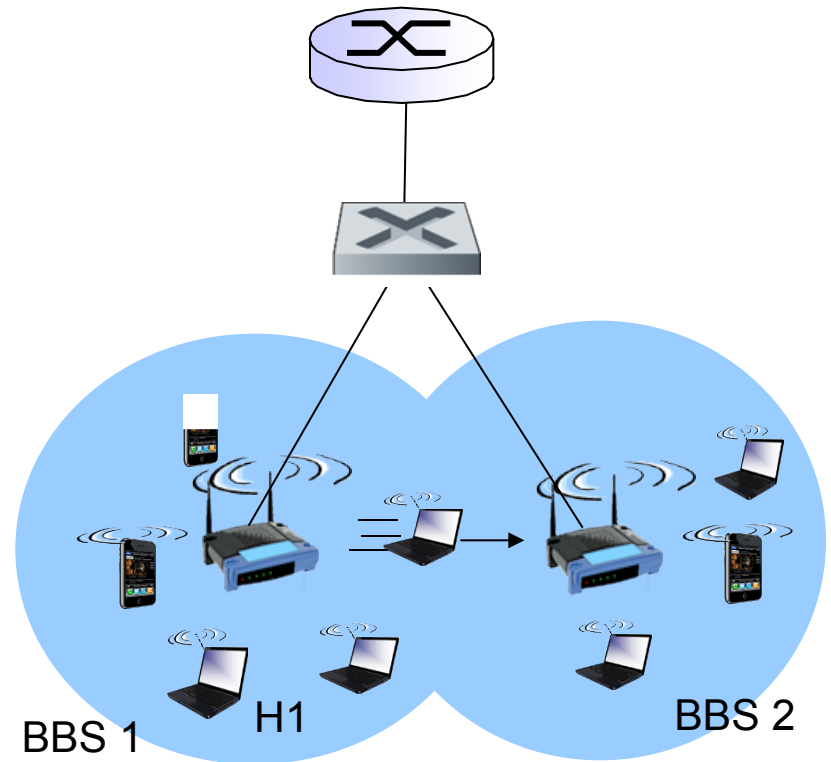


802.11: mobility within same subnet

- ❖ H1 remains in same IP subnet: IP address can remain same

is ass
with H1?

- self-learning (Ch. 5): switch will see frame from H1 and “remember” which switch port can be used to reach H1

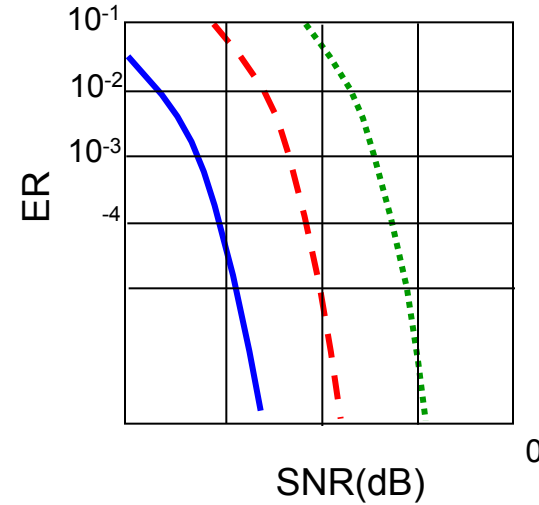
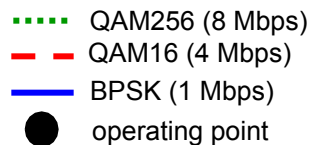


802.11: advanced capabilities

Rate adaptation

- ❖ base station, mobile dynamically change

modul
technique) as mobile
moves, SNR varies



1. SNR decreases, BER increase as node moves away from base station
2. When BER becomes too high, switch to lower transmission rate but with lower BER

802.11: advanced capabilities

power management

- ❖ node-to-AP: "I am going to sleep until next bea

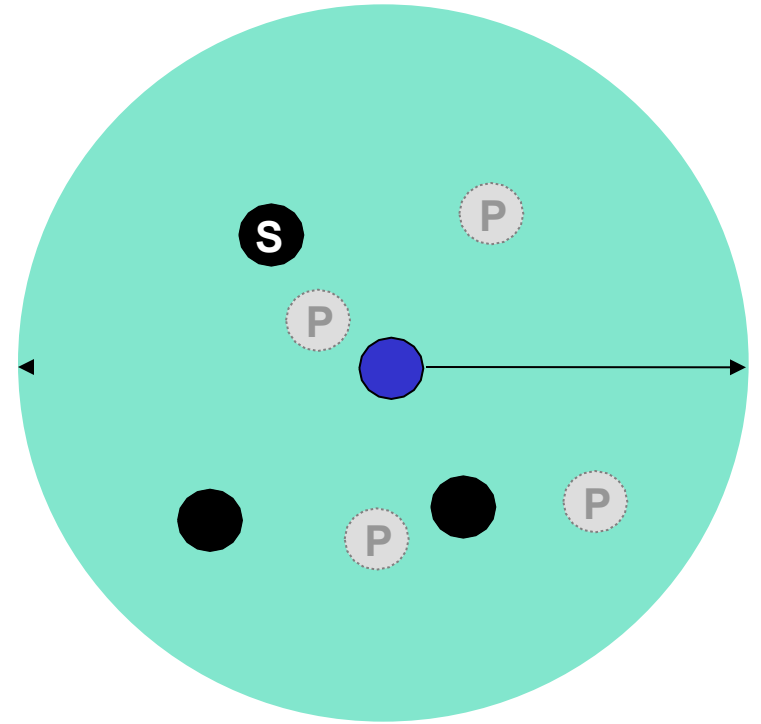
- node e

- ❖ beacon frame: contains list of mobiles with AP-to-mobile frames waiting to be sent

- node will stay awake if AP-to-mobile frames to be sent; otherwise sleep again until next beacon frame

802.15: personal area network

- ❖ less than 10 m diameter
- ❖ replacement for cables (mouse, keyboard, headphones)
- ❖ master
 - slaves request permission to send (to master)
 - master grants requests
- ❖ 802.15: evolved from Bluetooth specification
 - 2.4-2.5 GHz radio band
 - up to 721 kbps



- M** Master device
- S** Slave device
- P** Parked device (inactive)

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6.5 Principles: routing and

6.7 Handling mobility in cellular networks

6.8 Mobility and higher- layer protocols

6.9 Summary

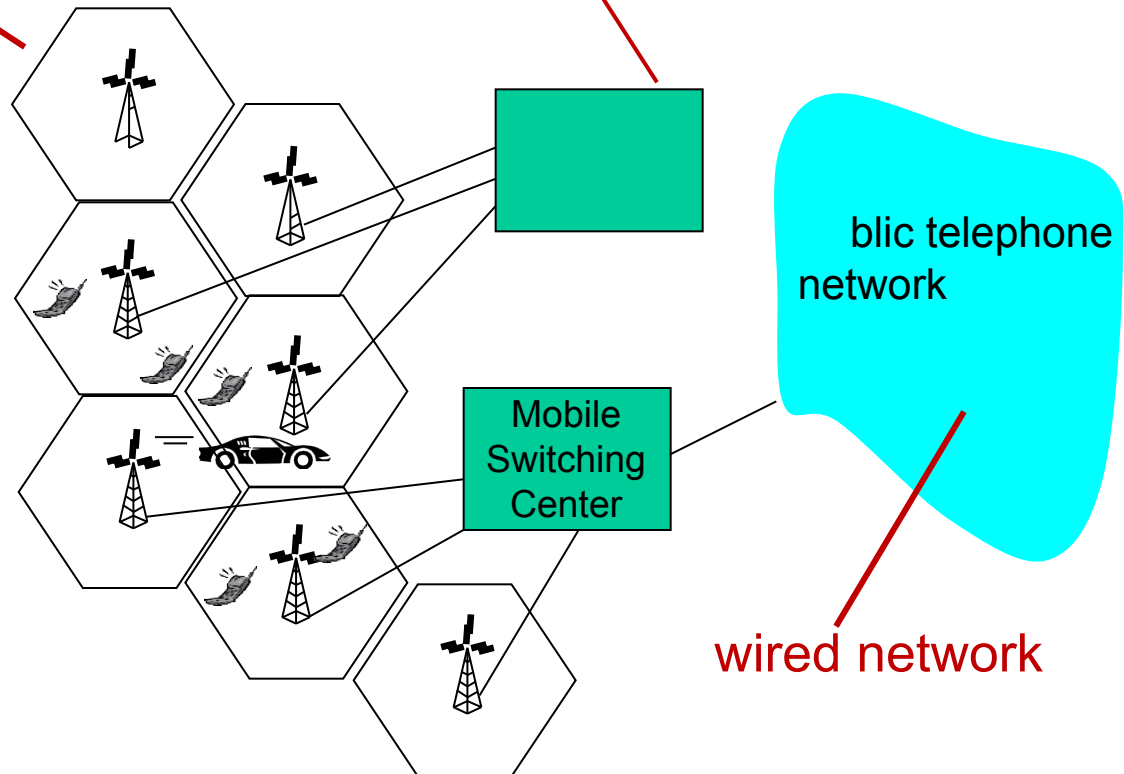
Components of cellular network architecture

cell

- ❖ covers geographic area
- ❖ *base station* (BS) analogous to 802.11 AP
- ❖ *mobile users* attach to network through BS
- ❖ *air-interface*: physical and link layer protocol between mobile and BS

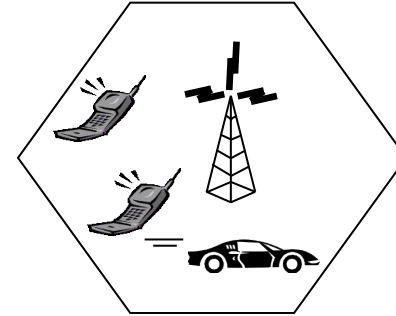
MSC

- ❖ connects cells to wired tel. net.
- ❖ manages call setup (more later!)
- ❖ handles mobility (more later!)



Cellular networks: the first hop

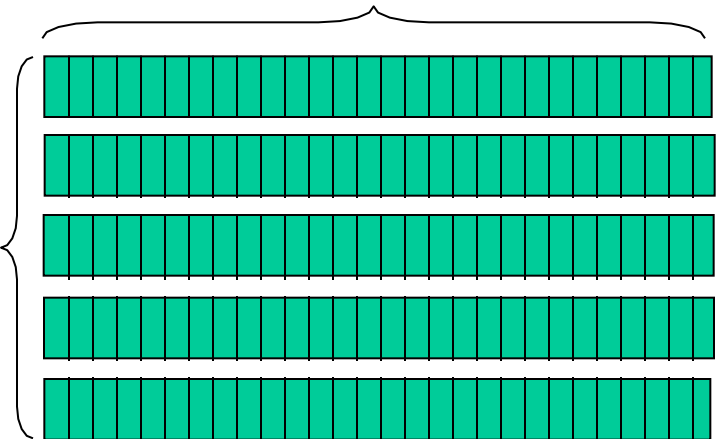
Two techniques for sharing mobile-to-BS radio spectrum



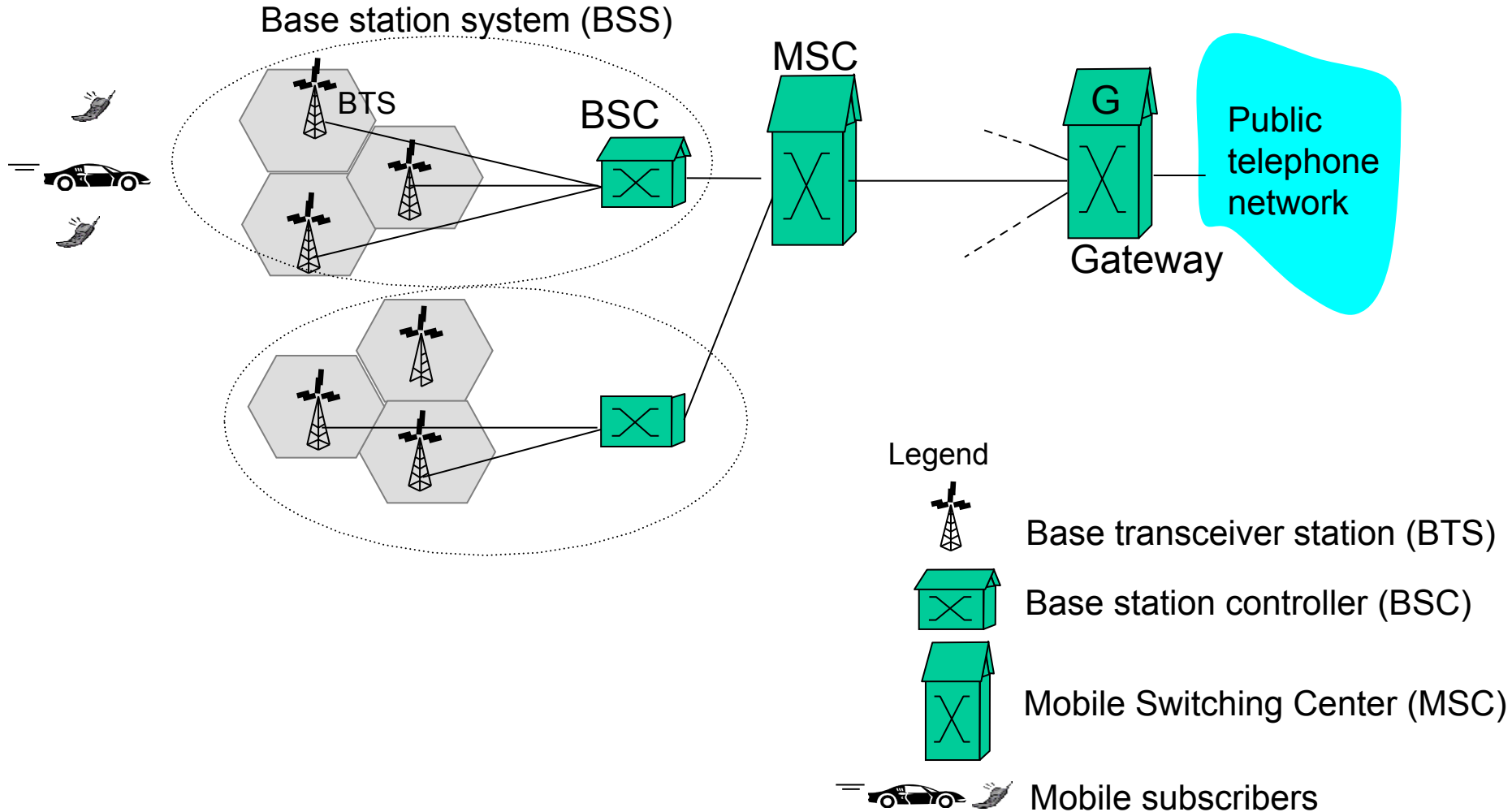
frequency
divide each channel into
time slots

- ❖ **CDMA:** code division multiple access

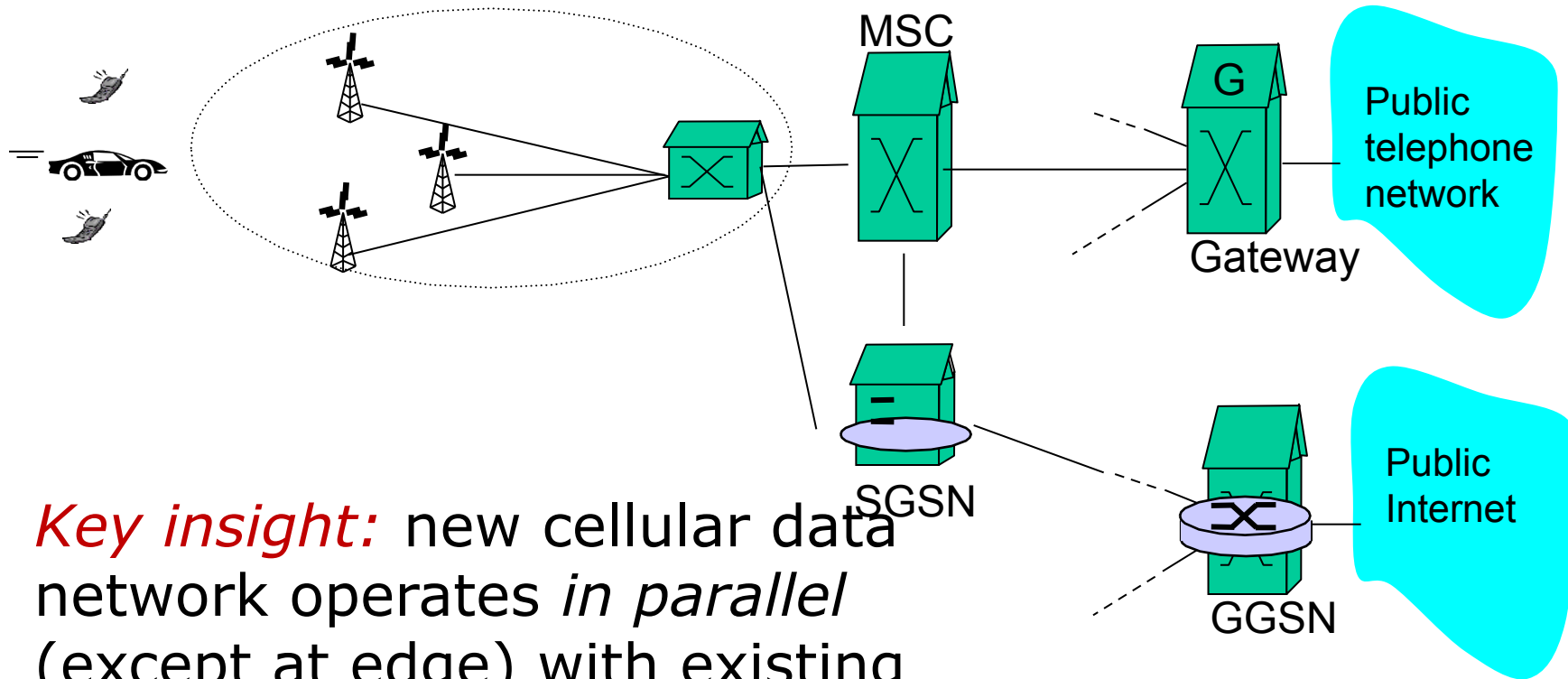
frequency
bands



2G (voice) network architecture



3G (voice+data) network architecture

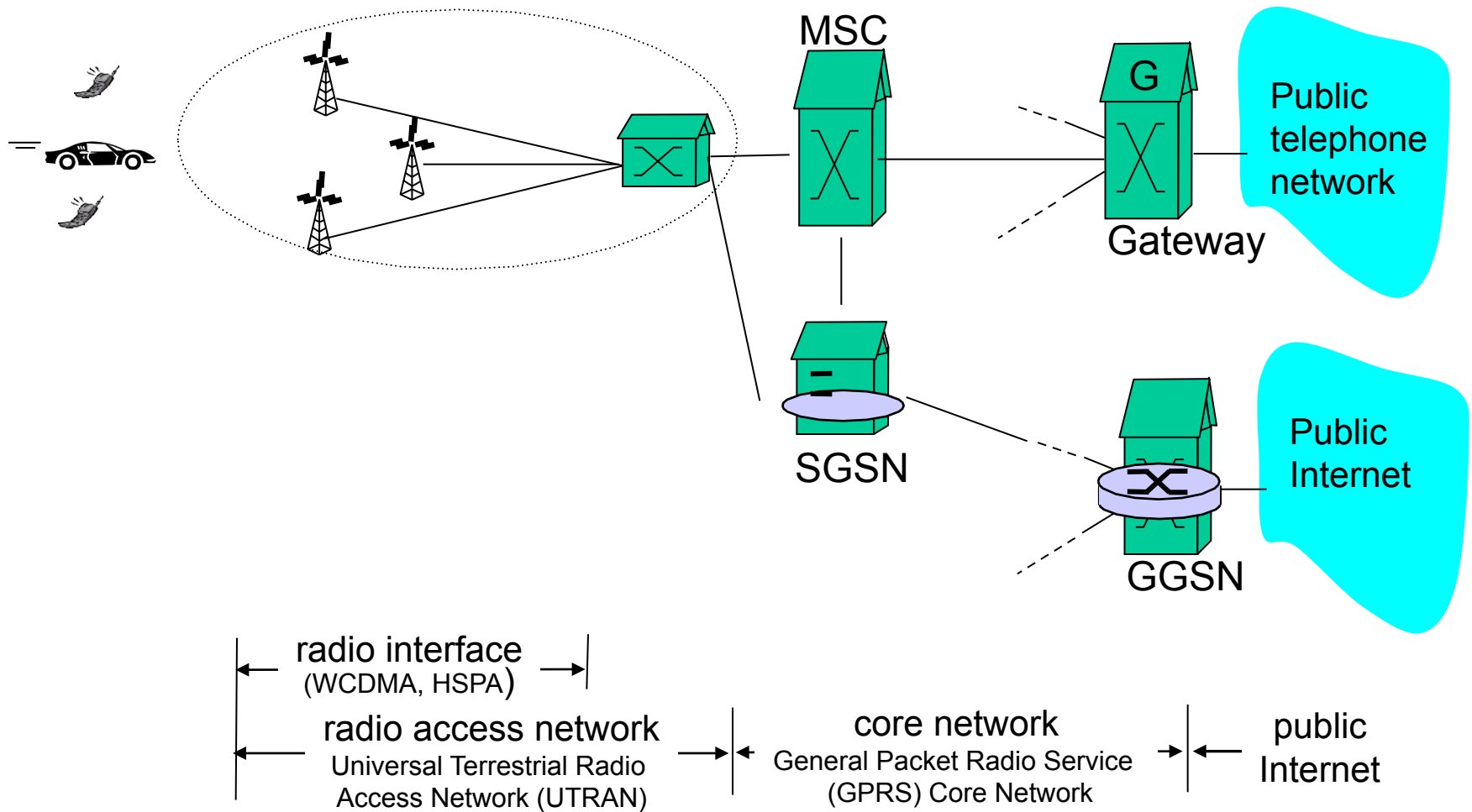


Key insight: new cellular data network operates *in parallel* (except at edge) with existing cellular voice network

- ❖ voice network unchanged in core
- ❖ data network operates in parallel



3G (voice+data) network architecture



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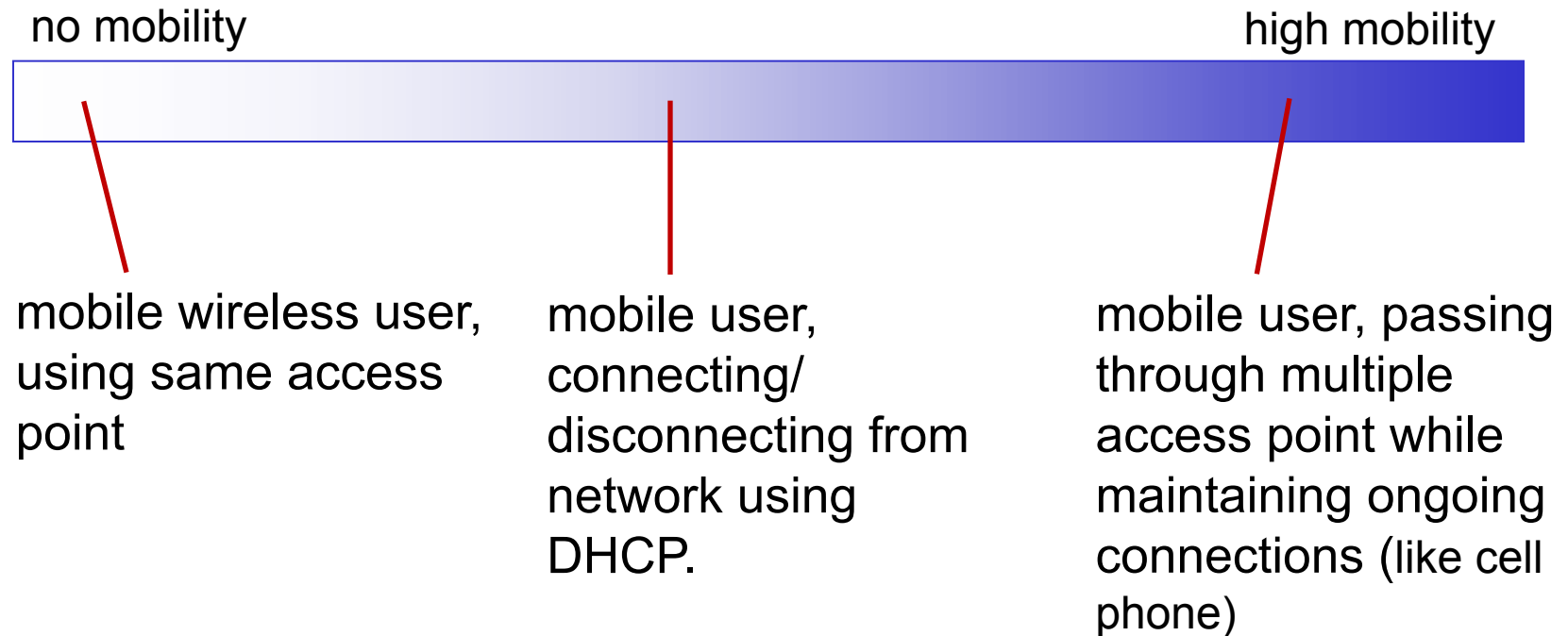
6.7 Handling mobility in cellular networks

6.8 Mobility and higher- layer protocols

6.9 Summary

What is mobility?

- ❖ spectrum of mobility, from the *network* perspective:

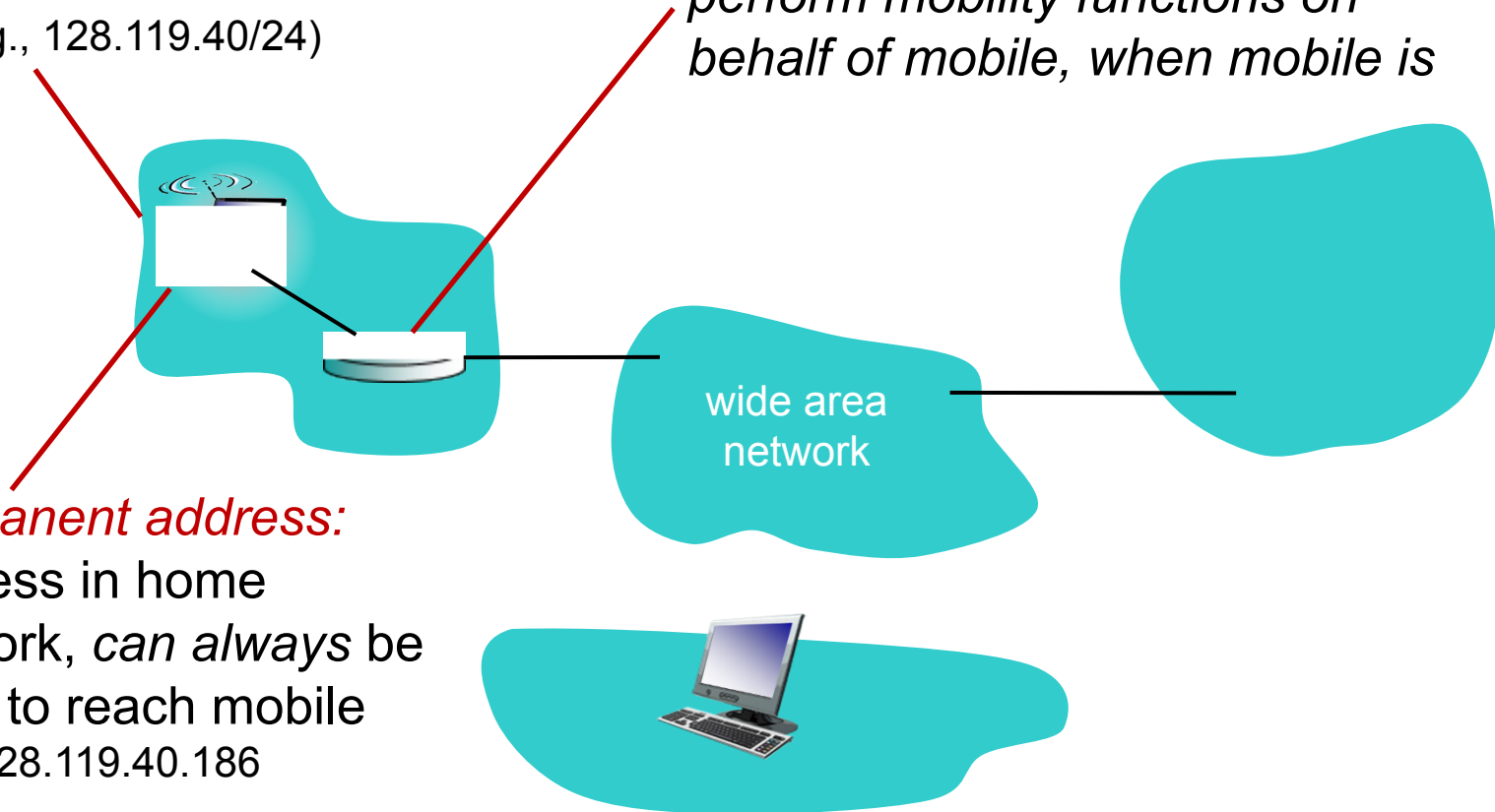


Mobility: vocabulary

home network: permanent
“home” of mobile
(e.g., 128.119.40/24)

home agent: entity that will
perform mobility functions on
behalf of mobile, when mobile is

permanent address:
address in home
network, *can always* be
used to reach mobile
e.g., 128.119.40.186



Mobility: more vocabulary

permanent address: remains constant (e.g., 128.119.40.186)

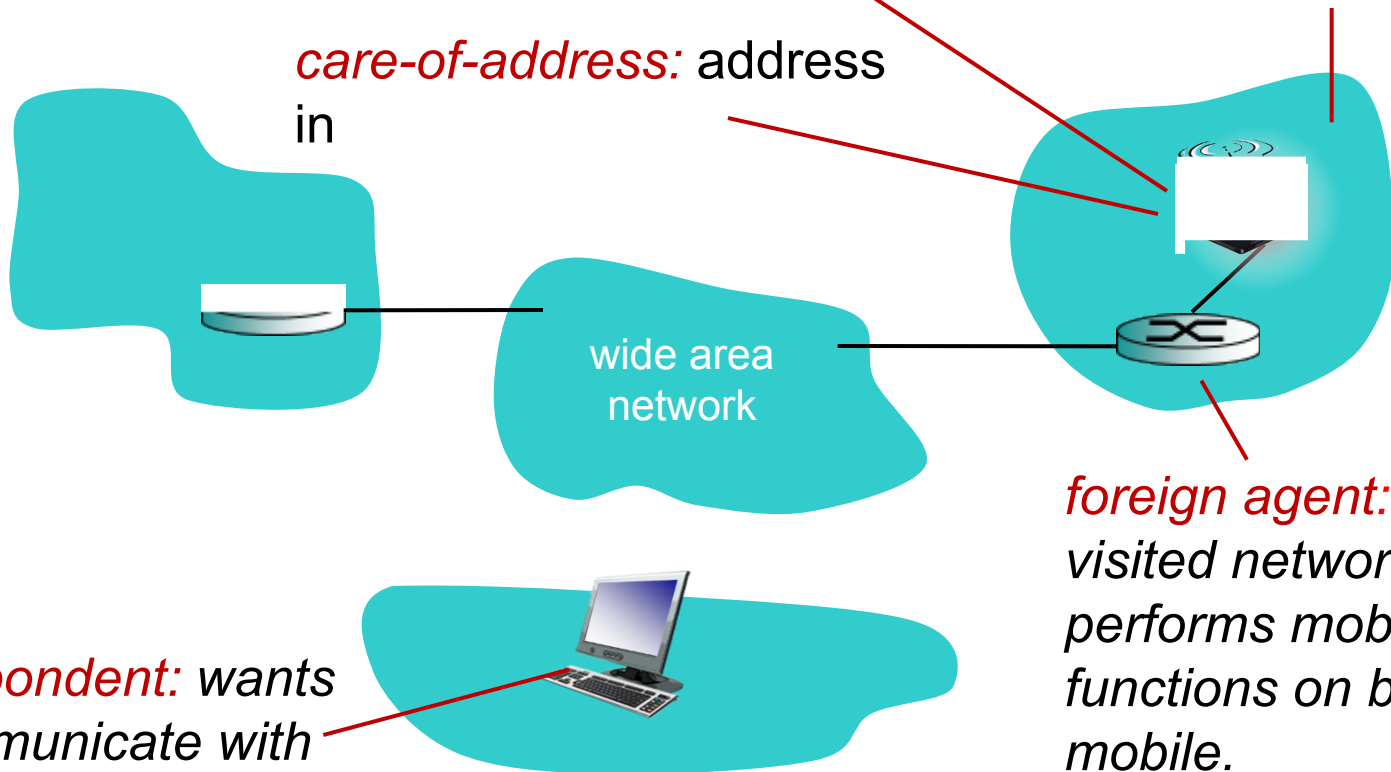
visited network: network in which mobile currently resides (e.g., 79.129.13/24)

care-of-address: address in

wide area network

foreign agent: entity in visited network that performs mobility functions on behalf of mobile.

correspondent: wants to communicate with mobile



How do *you* contact a mobile friend:

Consider friend frequently changing addresses, how do you find he

- ❖ search all phon

- ❖ call he
- ❖ expect her to let you know where he/she is?

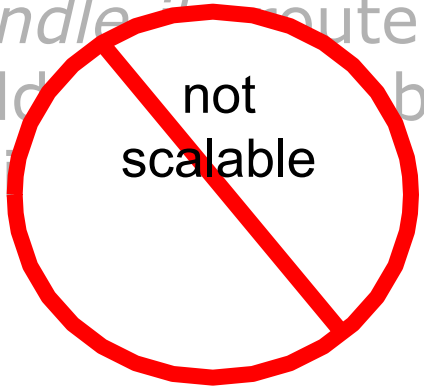
I wonder where Alice moved to?



Mobility: approaches

- ❖ *let routing handle it:* routers advertise permanent address of mobile-nodes-in-residence via usual routing protocol.
- no
- ❖ *let end-systems handle it:*
 - *indirect routing:* communication from correspondent to mobile goes through home agent, then forwarded to remote
 - *direct routing:* correspondent gets foreign address of mobile, sends directly to mobile

Mobility: approaches

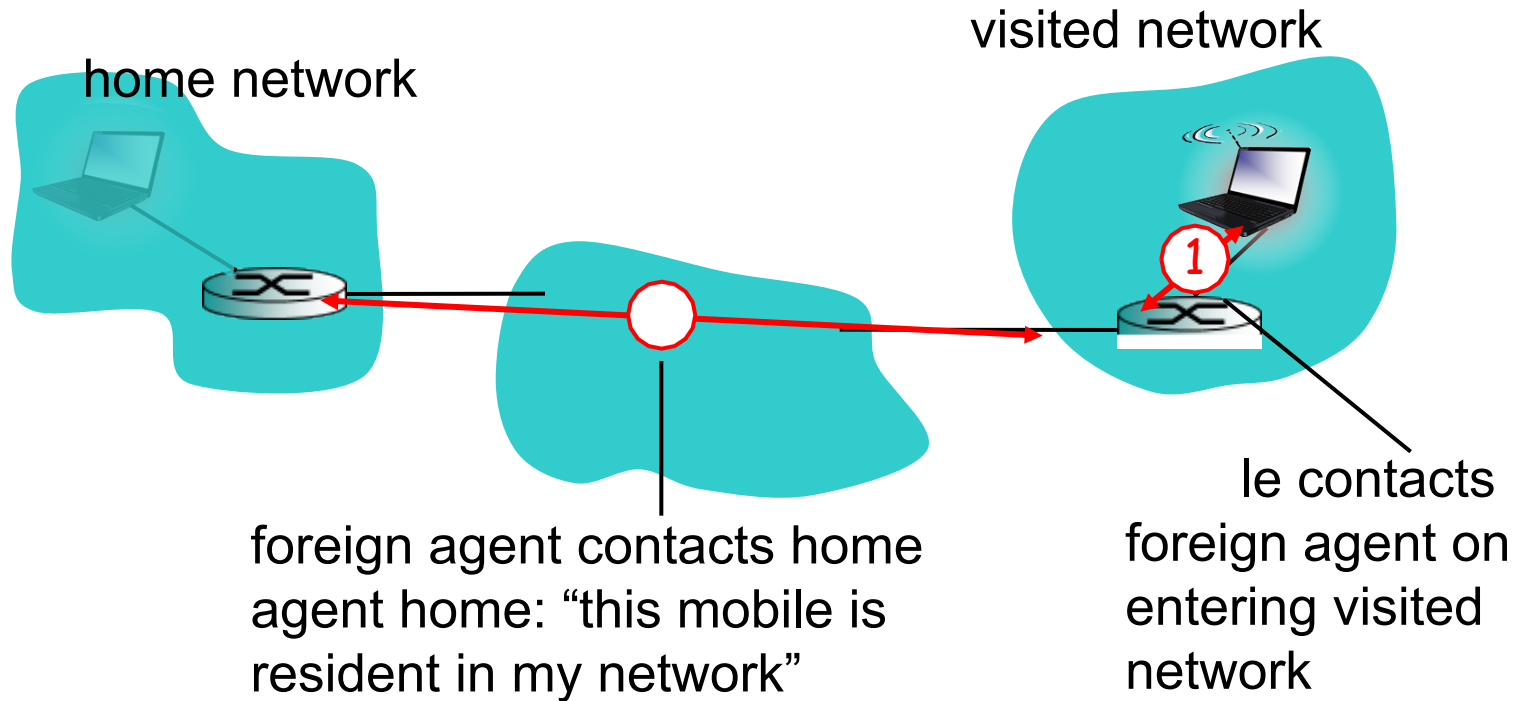
- ❖ *let routing handle it*: routers advertise permanent address of mobile-nodes-in-residence via usual routing protocols.  not scalable

- no

- ❖ *let end-systems handle it*:

- *indirect routing*: communication from correspondent to mobile goes through home agent, then forwarded to remote
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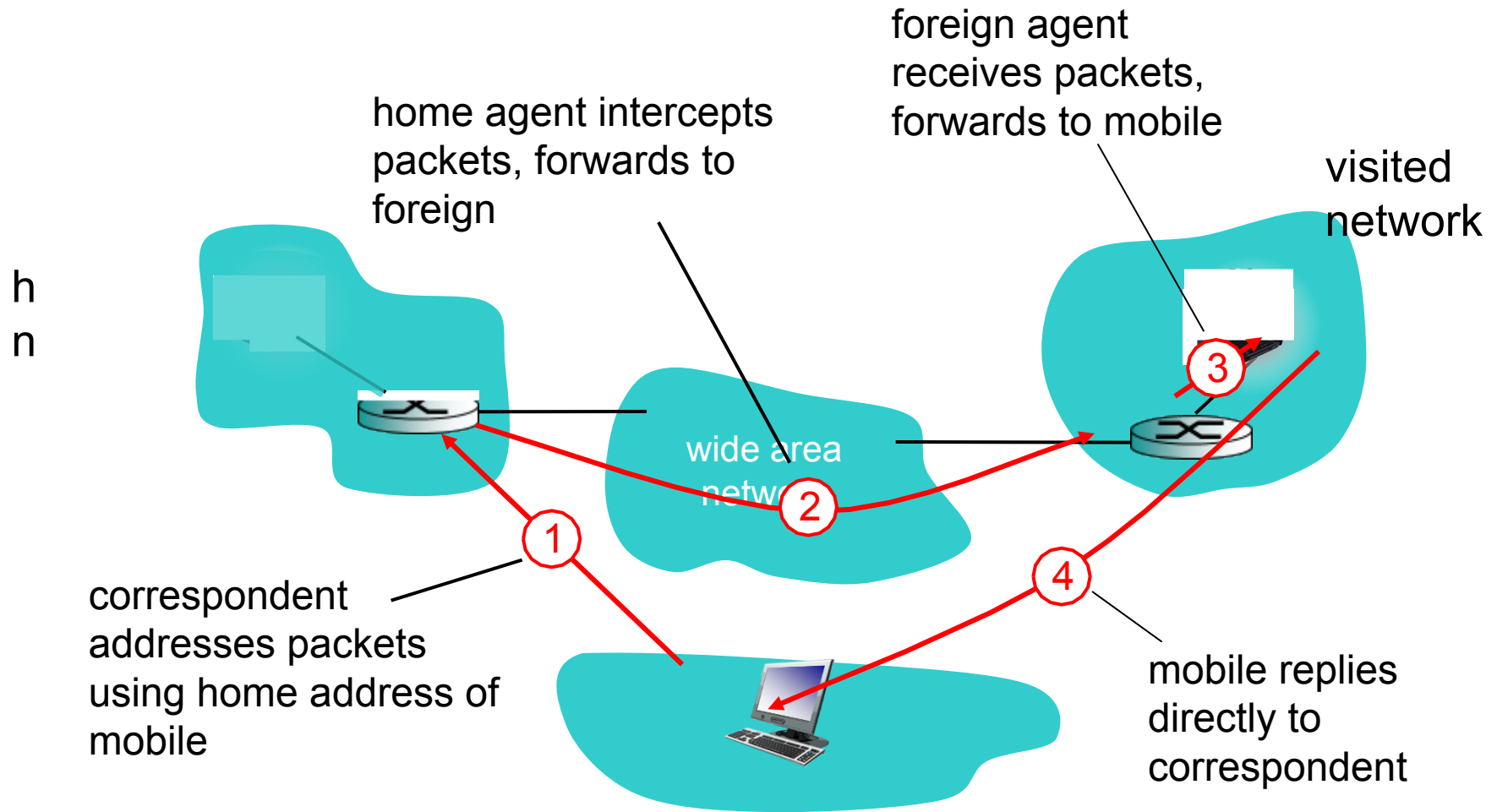
Mobility: registration



end result:

- ❖ foreign agent knows about mobile
- ❖ home agent knows location of mobile

Mobility via indirect routing

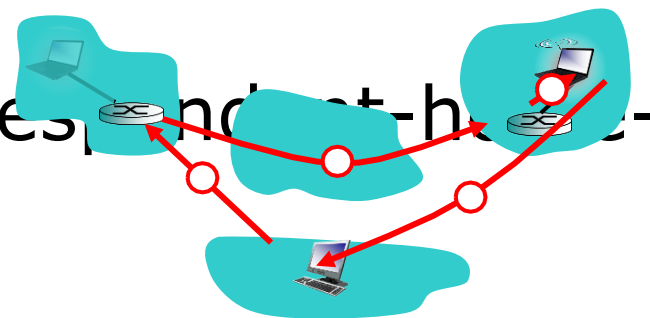


Indirect Routing: comments

- ❖ mobile uses two addresses:
 - **permanent address:** used by correspondent (hence mobile location is *transparent* to correspondent)

for

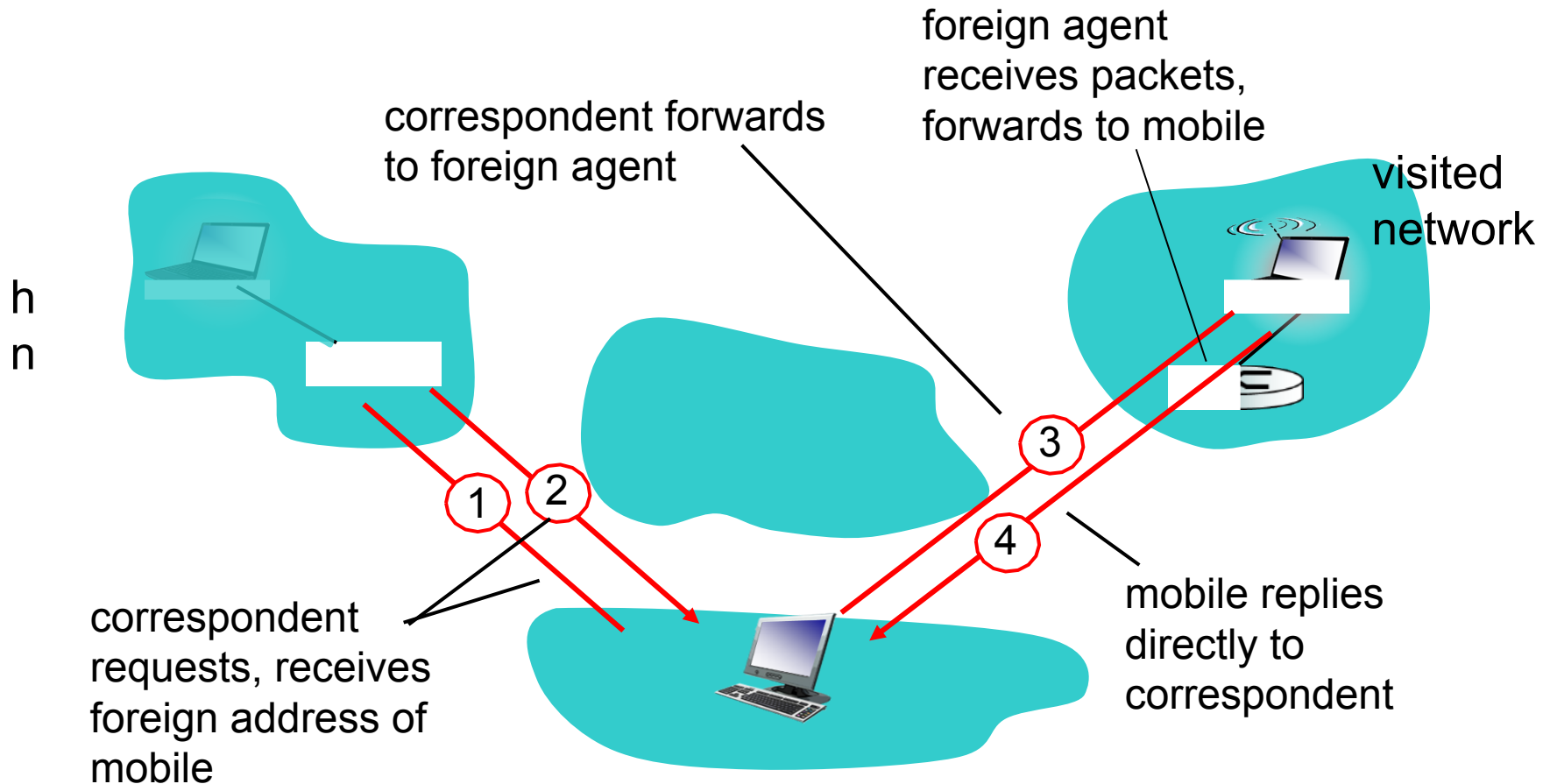
- ❖ foreign agent functions may be done by mobile itself
- ❖ **triangle routing:** correspondent-host-network-mobile
 - inefficient when correspondent, mobile are in same network



Indirect routing: moving between networks

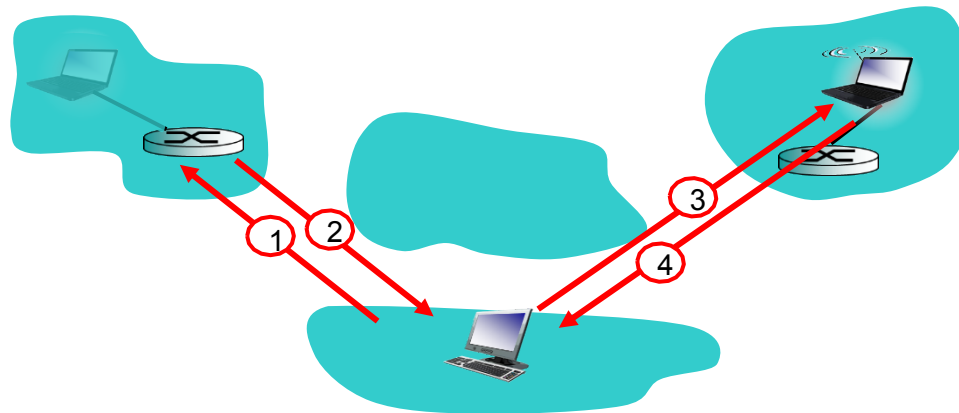
- ❖ suppose mobile user moves to another network
 - registers with home agent
 - home agent updates care-of-address for mobile
 - packets continue to be forwarded to mobile (but with new care-of-address)
- ❖ mobility, changing foreign networks transparent: *on going connections can be maintained!*

Mobility via direct routing



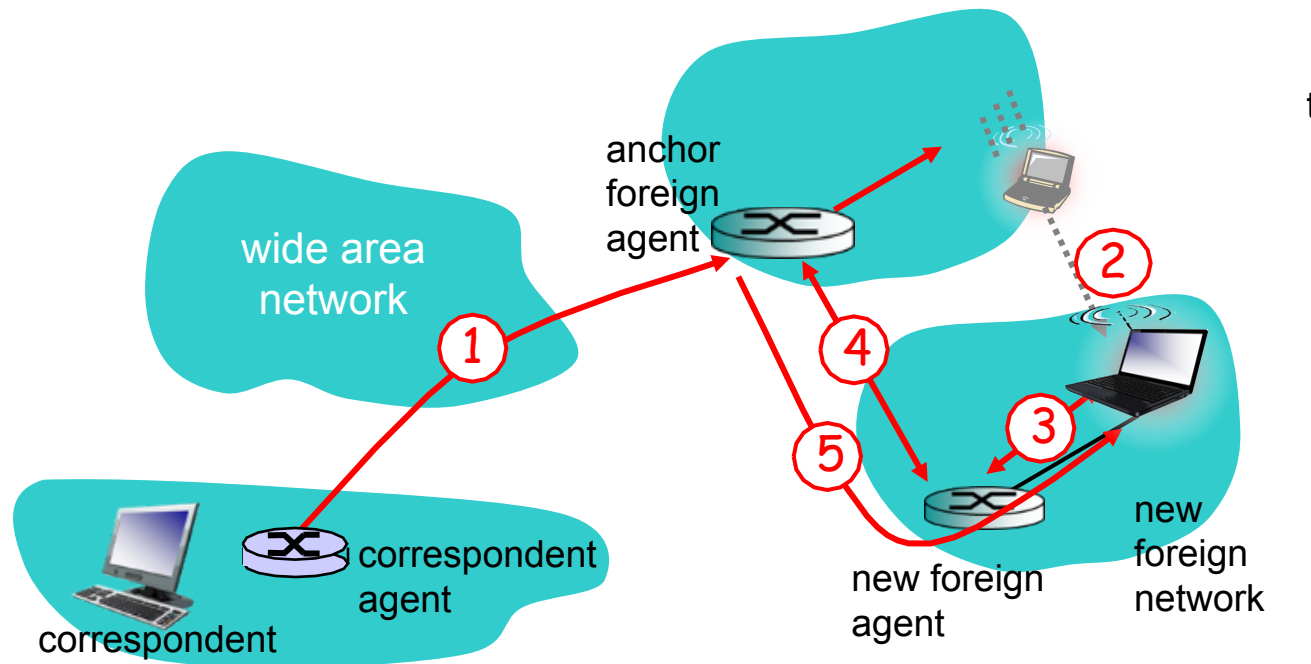
Mobility via direct routing: ~~comments~~

- ❖ overcome triangle routing problem
- ❖ *non-transparent to correspondent:*
corresponde re-of-address



Accommodating mobility with direct routing

- ❖ anchor foreign agent: FA in first visited network
- ❖ data always routed first to anchor FA
- ❖ when mobile A arrives



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Mobile IP

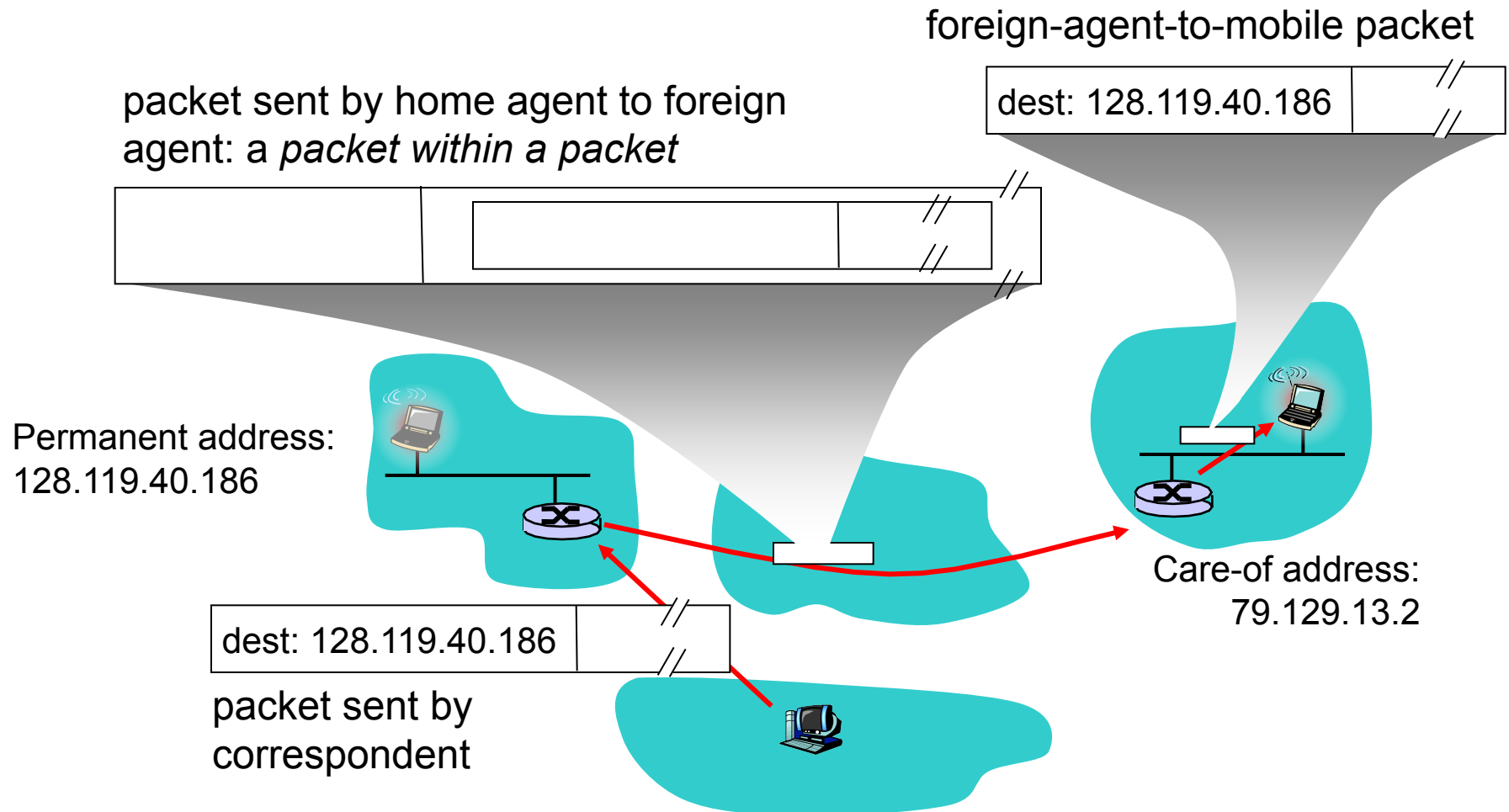
- ❖ RFC 3344
- ❖ has many features we've seen:

enc

)

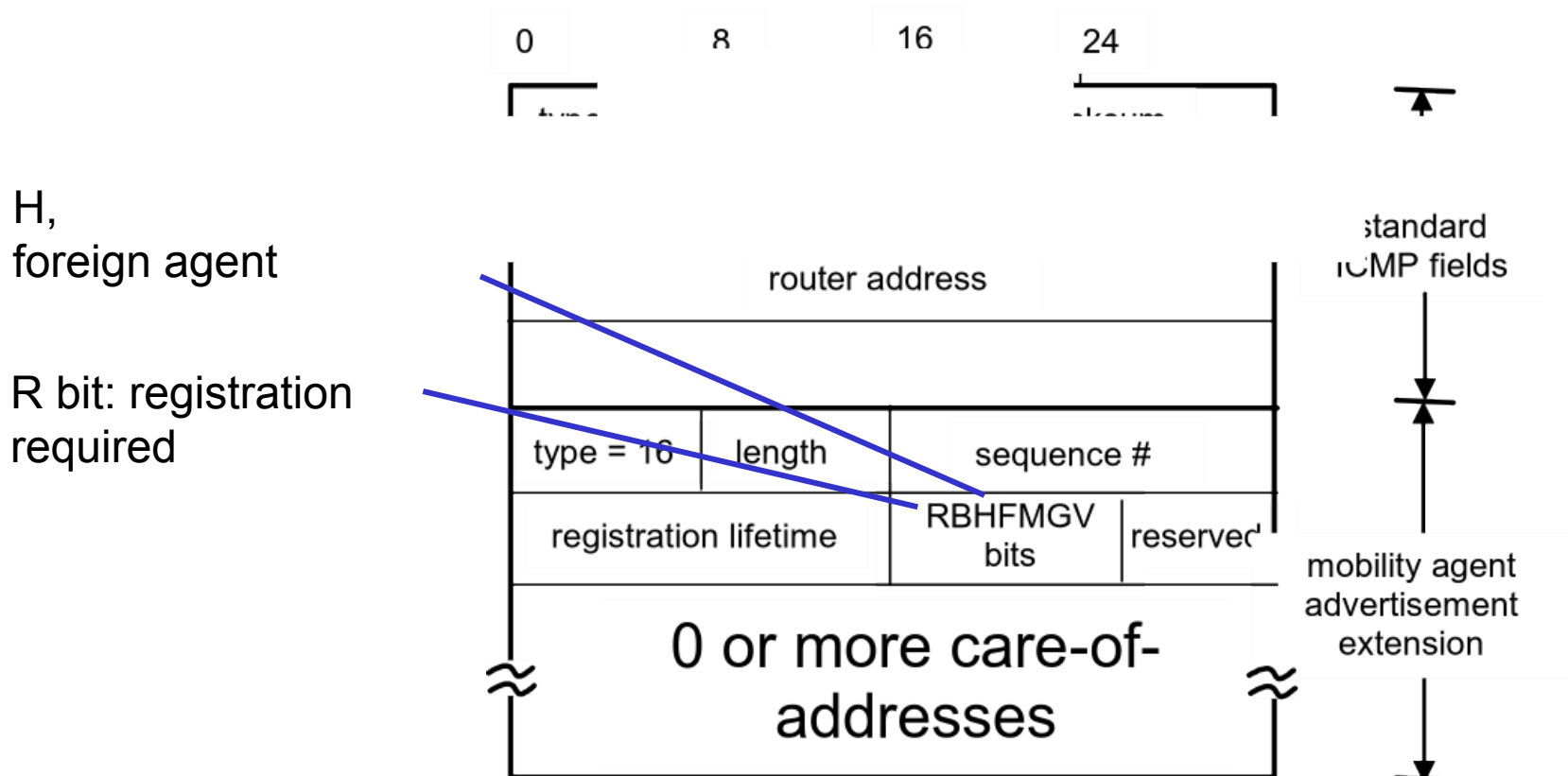
- ❖ three components to standard:
 - indirect routing of datagrams
 - agent discovery
 - registration with home agent

Mobile IP: indirect routing

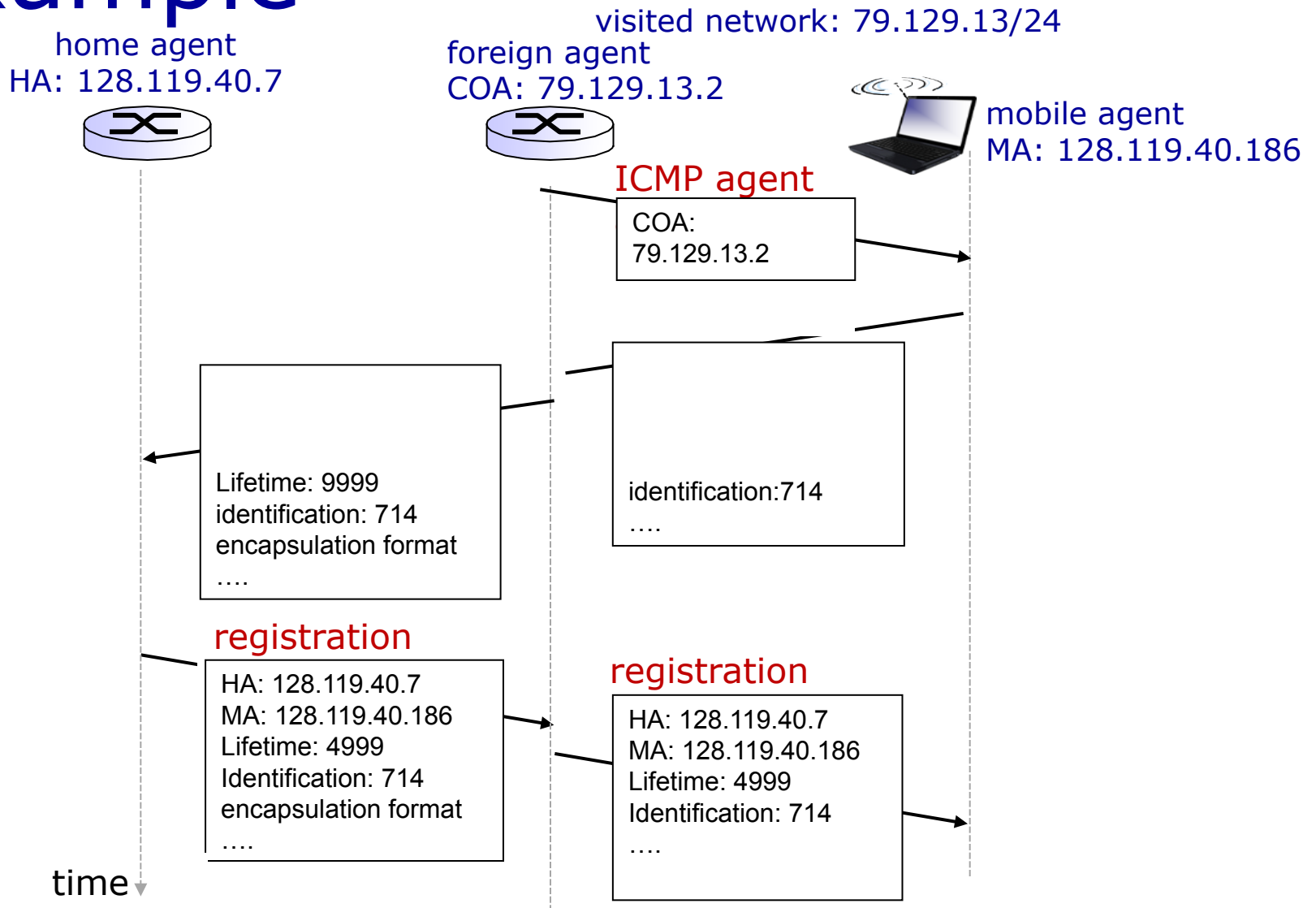


Mobile IP: agent discovery

- ❖ *agent advertisement*: foreign/home agents advertise service by broadcasting ICMP messages (typefield = 9)

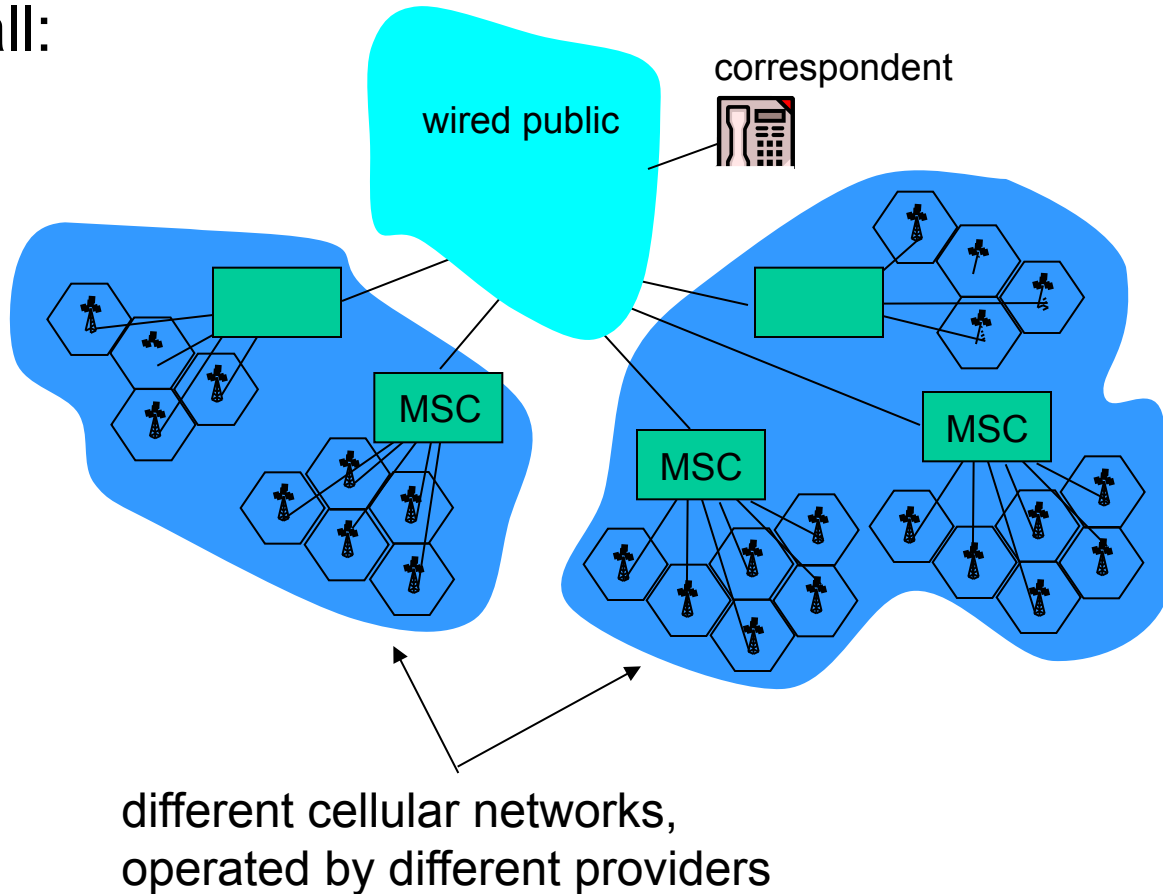


Mobile IP: registration example



Components of cellular network architecture

recall:



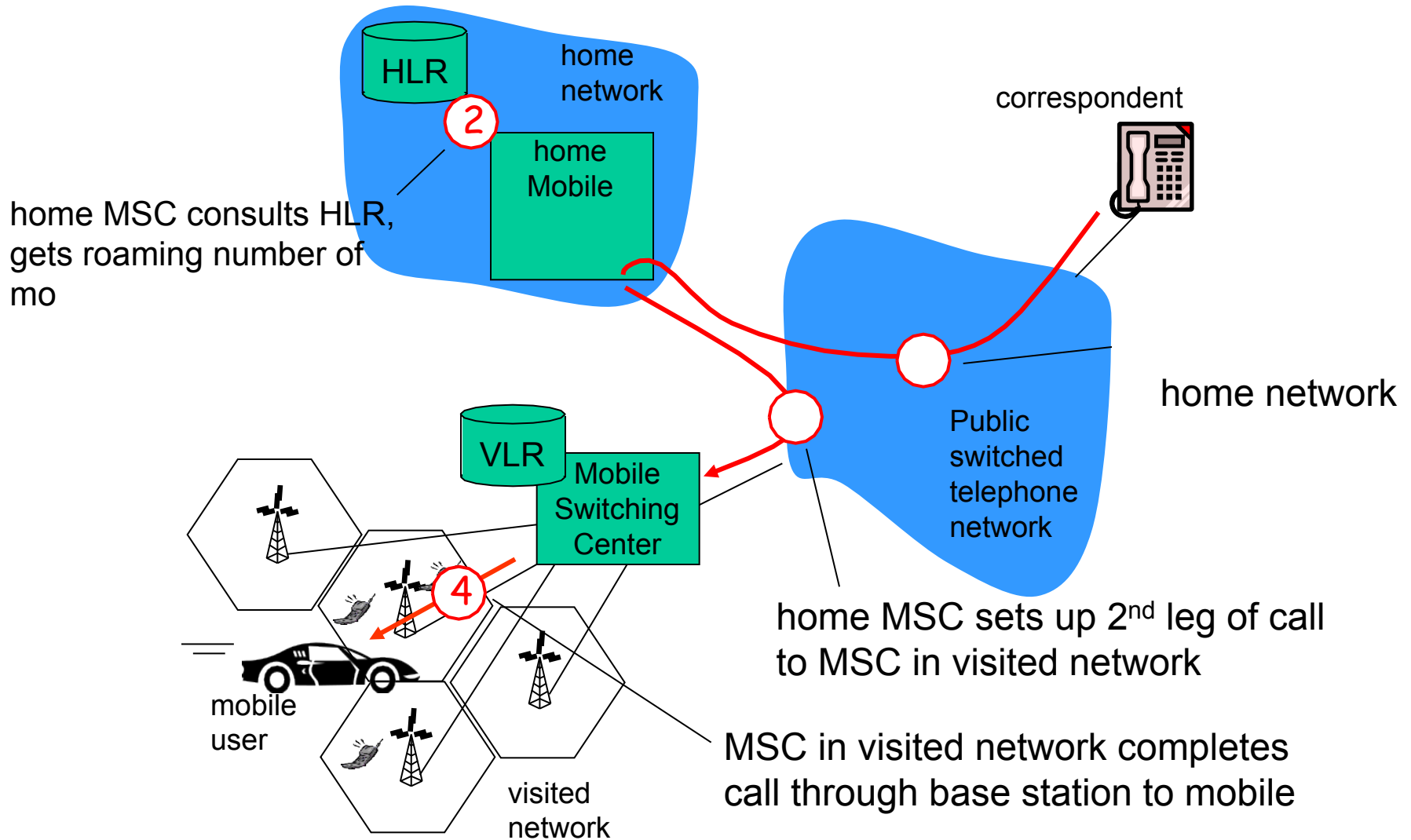
Handling mobility in cellular networks

- ❖ *home network*: network of cellular provider you subscribe to (e.g., Sprint PCS, Verizon)

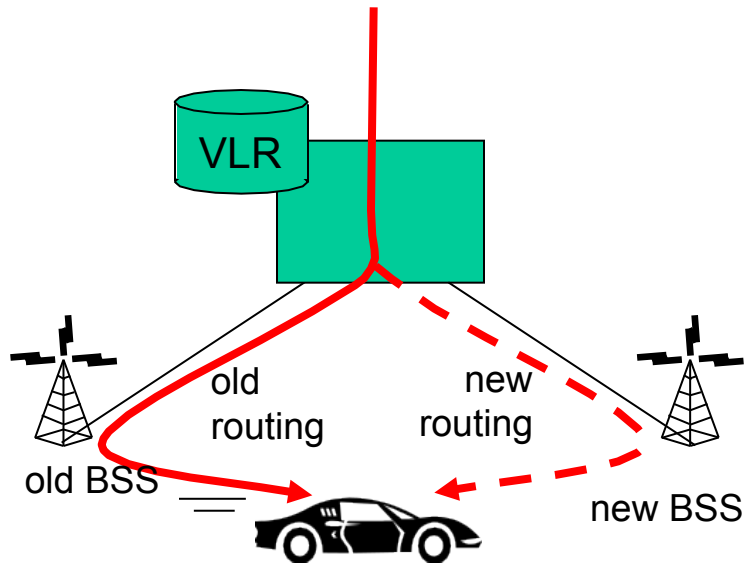
ph
preferences, billing), information about current location (could be in another network)

- ❖ *visited network*: network in which mobile currently resides
 - *visitor location register (VLR)*: database with entry for each user currently in network

GSM: indirect routing to mobile



GSM: handoff with common MSC

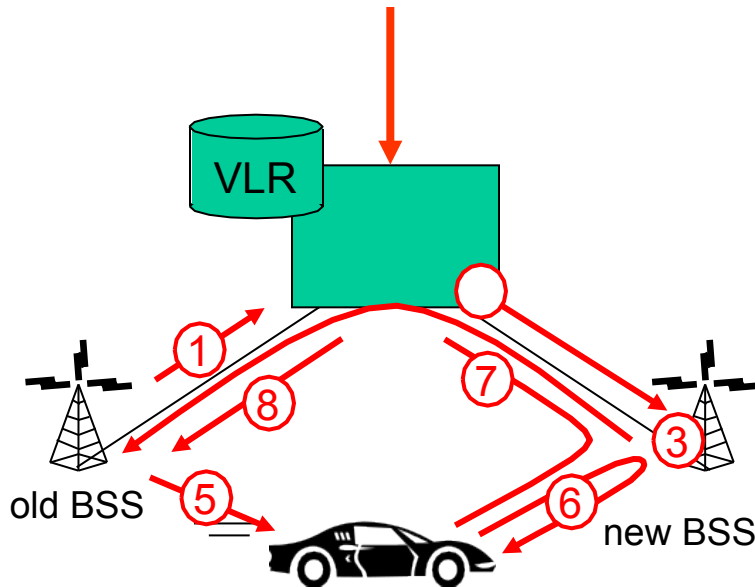


- ❖ *handoff goal:* route call via new base station (at interruption)

connectivity, less battery drain)

- load balance: free up channel in current BSS
 - GSM doesn't mandate why to perform handoff (policy), only how (mechanism)
- ❖ handoff initiated by old BSS

GSM: handoff with common MSC



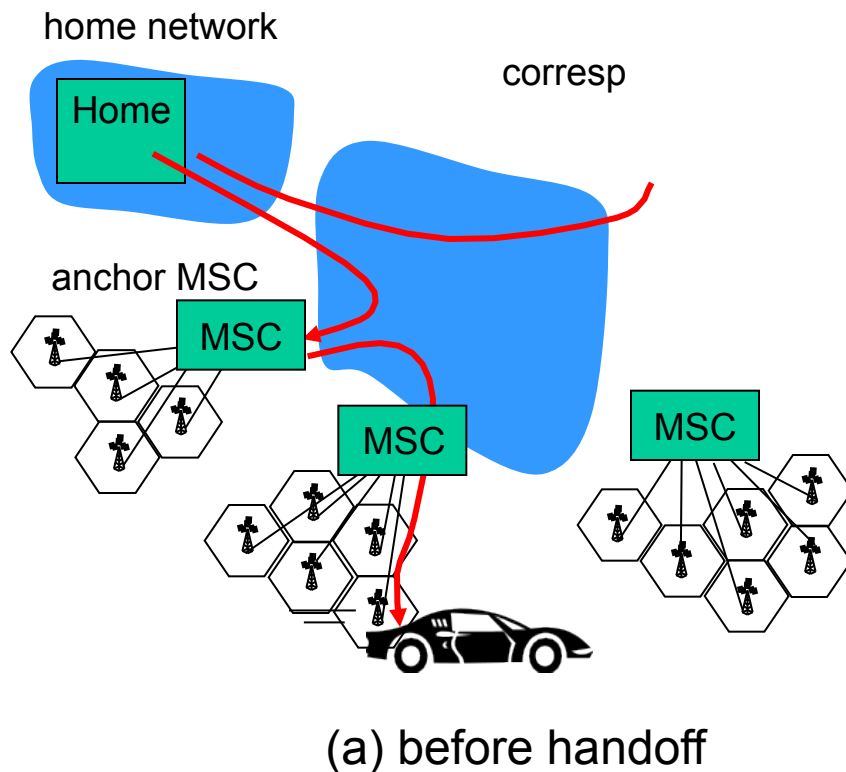
1. old BSS informs MSC of impending handoff, provides list of 1+ new BSSs
2. MSC sets up path (allocates resources)

r

Id BSS: ready

5. old BSS tells mobile: perform handoff to new BSS
6. mobile, new BSS signal to activate new channel
7. mobile signals via new BSS to MSC: handoff complete. MSC reroutes call
8. MSC-old-BSS resources released

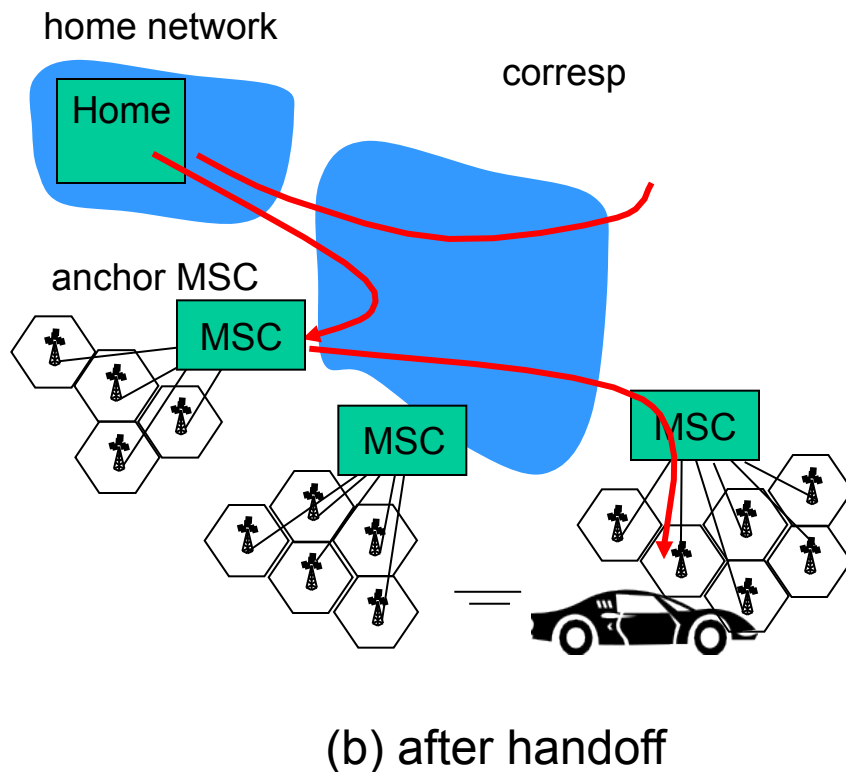
GSM: handoff between MSCs



- ❖ *anchor MSC*: first MSC visited during call
It remains routed C

- ain as mobile moves to new MSC
- ❖ optional path minimization step to shorten multi-MSC chain

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Mobility: GSM versus Mobile IP

GSM element	Comment on GSM element	Mobile IP element
Home system	Network to which mobile user's permanent phone number belongs	Home network
Gateway Mobile Switching Center, or "home MSC". Home	Home MSC: point of contact to obtain routable address in home permanent phone	Home agent
Visited Syst	mobile user is currently residing	Visited network
Visited Mobile services Switching Center. Visitor Location Record (VLR)	Visited MSC: responsible for setting up calls to/from mobile nodes in cells associated with MSC. VLR: temporary database entry in visited system, containing subscription information for each visiting mobile user	Foreign agent
Mobile Station Roaming Number (MSRN), or "roaming number"	Routable address for telephone call segment between home MSC and visited MSC, visible to neither the mobile nor the correspondent.	Care-of-address

Wireless, mobility: impact on higher layer protocols

❖ logically, impact *should* be minimal ...

- best effort service model remains unchanged
- TCP and UDP over wireless,



- packet loss/delay due to bit-errors (discarded packets, delays for link-layer retransmissions), and handoff
- TCP interprets loss as congestion, will decrease congestion window un-necessarily
- delay impairments for real-time traffic
- limited bandwidth of wireless links

Chapter 6 summary

Wireless

- ❖ wireless links:
 - capacity, distance
- ❖ IEEE 802.11 (Wi-Fi)
 - CSMA/CA reflects wireless channel characteristics
- ❖ cellular access
 - architecture
 - standards (e.g., GSM, 3G, 4G LTE)

Mobility

- ❖ principles: addressing, routing to mobile
 - care-of-addresses
- ❖ case studies
 - mobile IP
 - mobility in GSM
- ❖ impact on higher-layer protocols